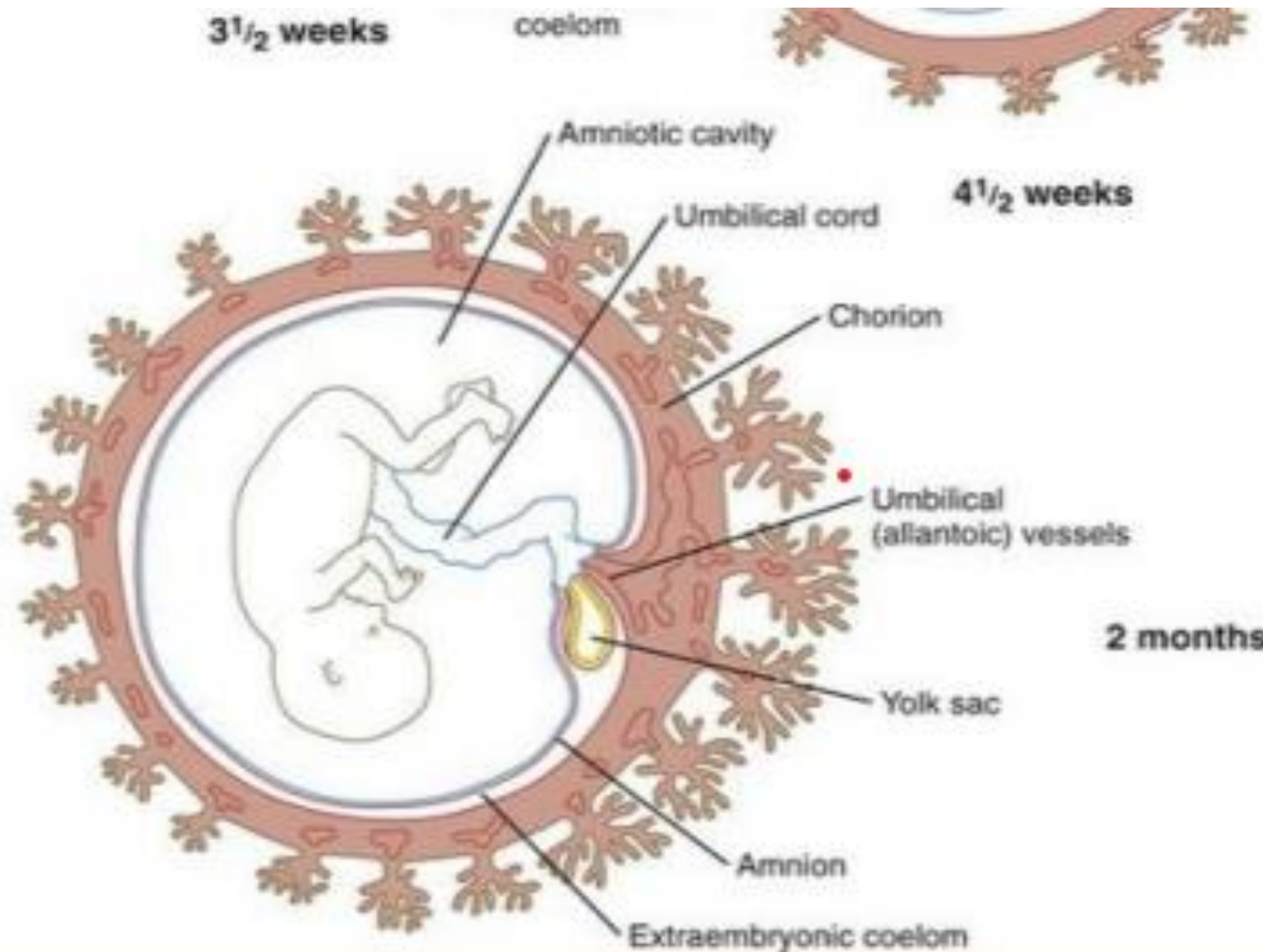


Placenta and Twinning

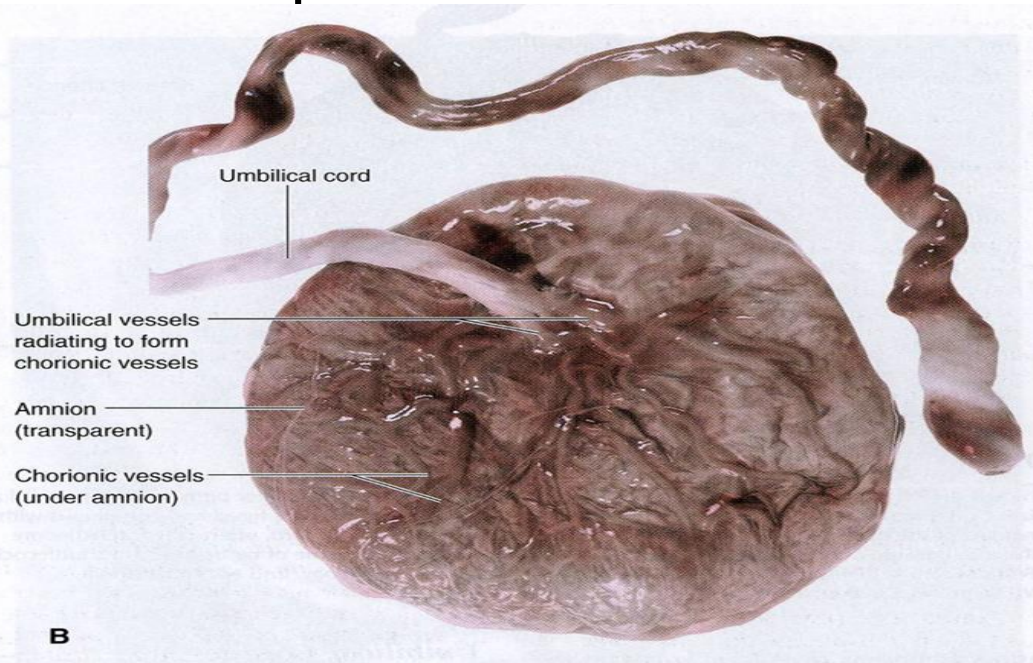
D . Ndhlovu-Chikwanda

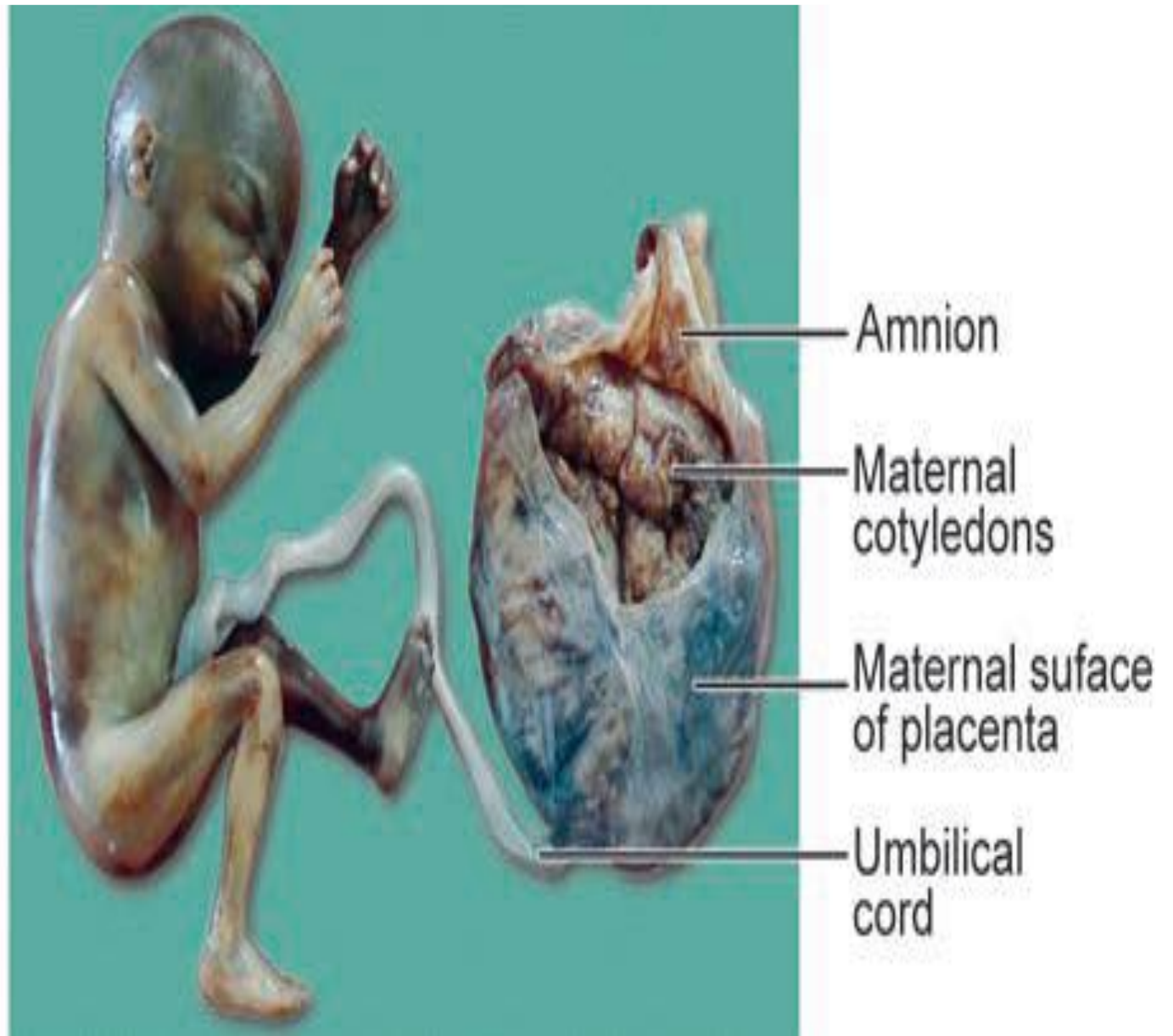
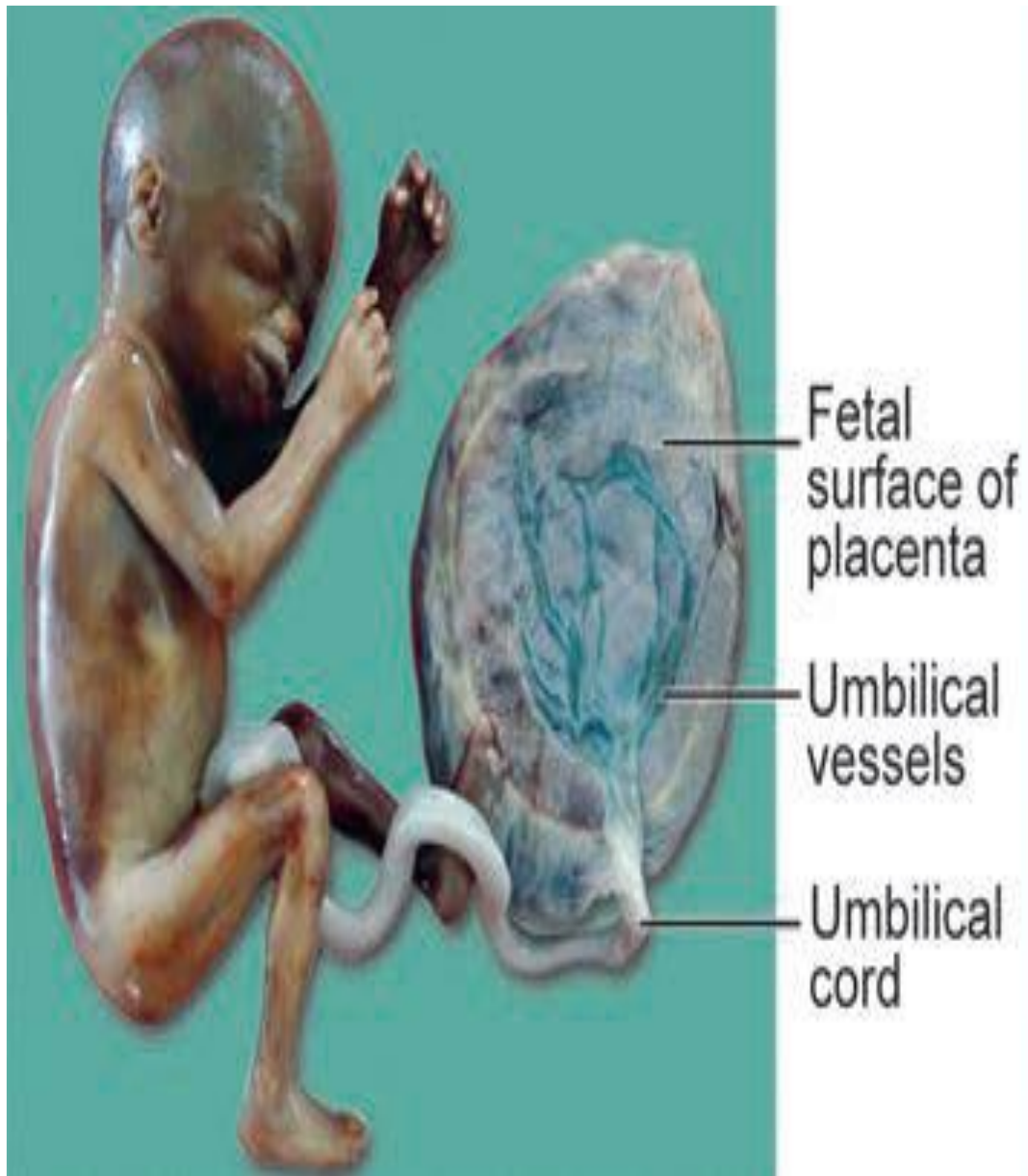
- **Placenta**
- Connects growing embryo/fetus with the wall of uterus
- primary site of nutrient & gas exchange between the mother & fetus
- **fetomaternal organ** that has 2 components:
- **fetal part** develops from the chorionic sac
- **maternal part** is derived from the endometrium
- <https://youtu.be/61dGxLw4iQs>
- <https://youtu.be/ZWZv0lwrQBk>

- **FORMATION OF PLACENTA**
- is a fetomaternal organ
- circular or disc-shaped organ weighs 500 g
- presents **2 surfaces & 2 structural components**



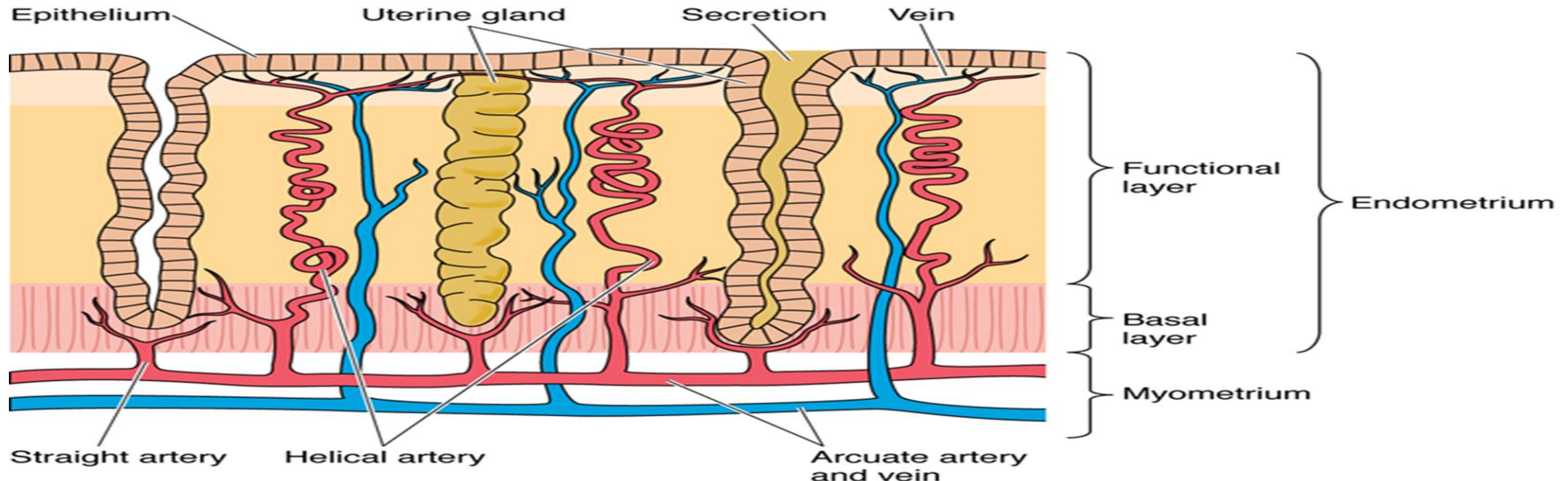
- **Surfaces**
- **Maternal surface**- irregular & divided into 15–20 lobules known as **maternal cotyledons**.
- **Fetal surface**- smooth & covered with amnion & umbilical cord is attached at or near the center of this surface
- **Structural components:**
- Maternal component derived from the ***decidua basalis*** or ***decidual plate***
- Fetal component derived from ***chorion frondosum*** or ***chorionic plate***



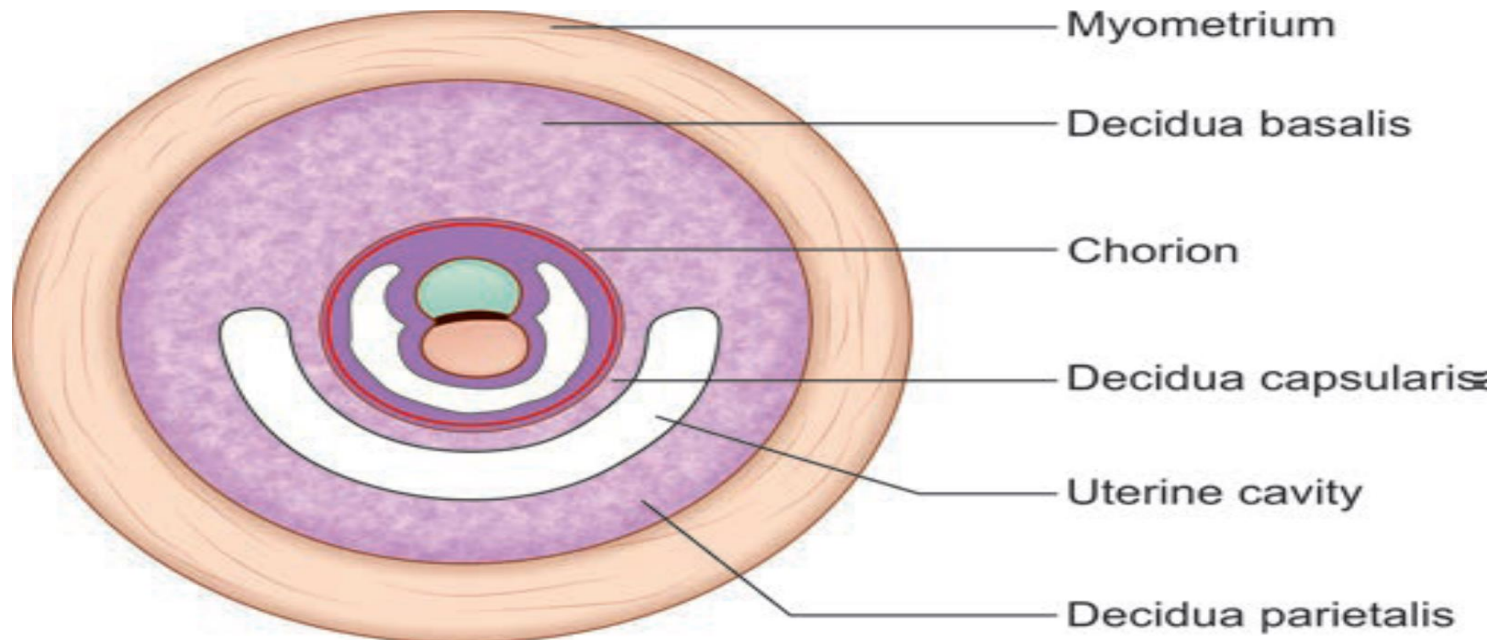


- **Decidua**
- functional stratum (stratum compactum) of uterine endometrium after the implantation of blastocyst
- separates and falls off during childbirth
- ***Decidual reaction:***
- Due to the higher levels of maternal progesterone and the hCG
- stromal cells enlarge, become vacuolated and filled with glycogen and lipids (decidual transformation)
- cells are called ***decidual cells***

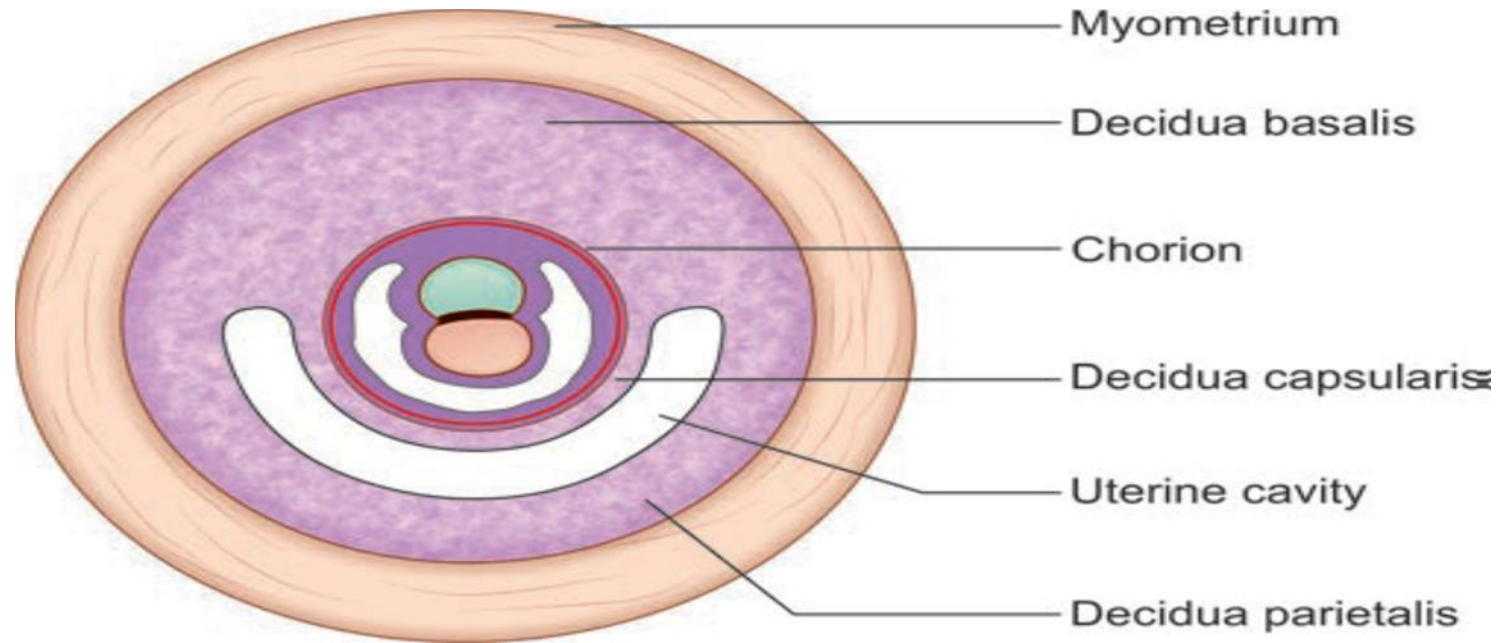
- intercellular substance increases, gives edematous appearance
- change in the endometrial stroma is called ***decidual reaction***
- glycogen and lipids provide nutrition to the embryo until the placenta takes over this function
- saw-toothed appearance of endometrial glands increases and the blood vessels of endometrium become more tortuous
- decidual reaction is a defensive mechanism to protect the endometrium



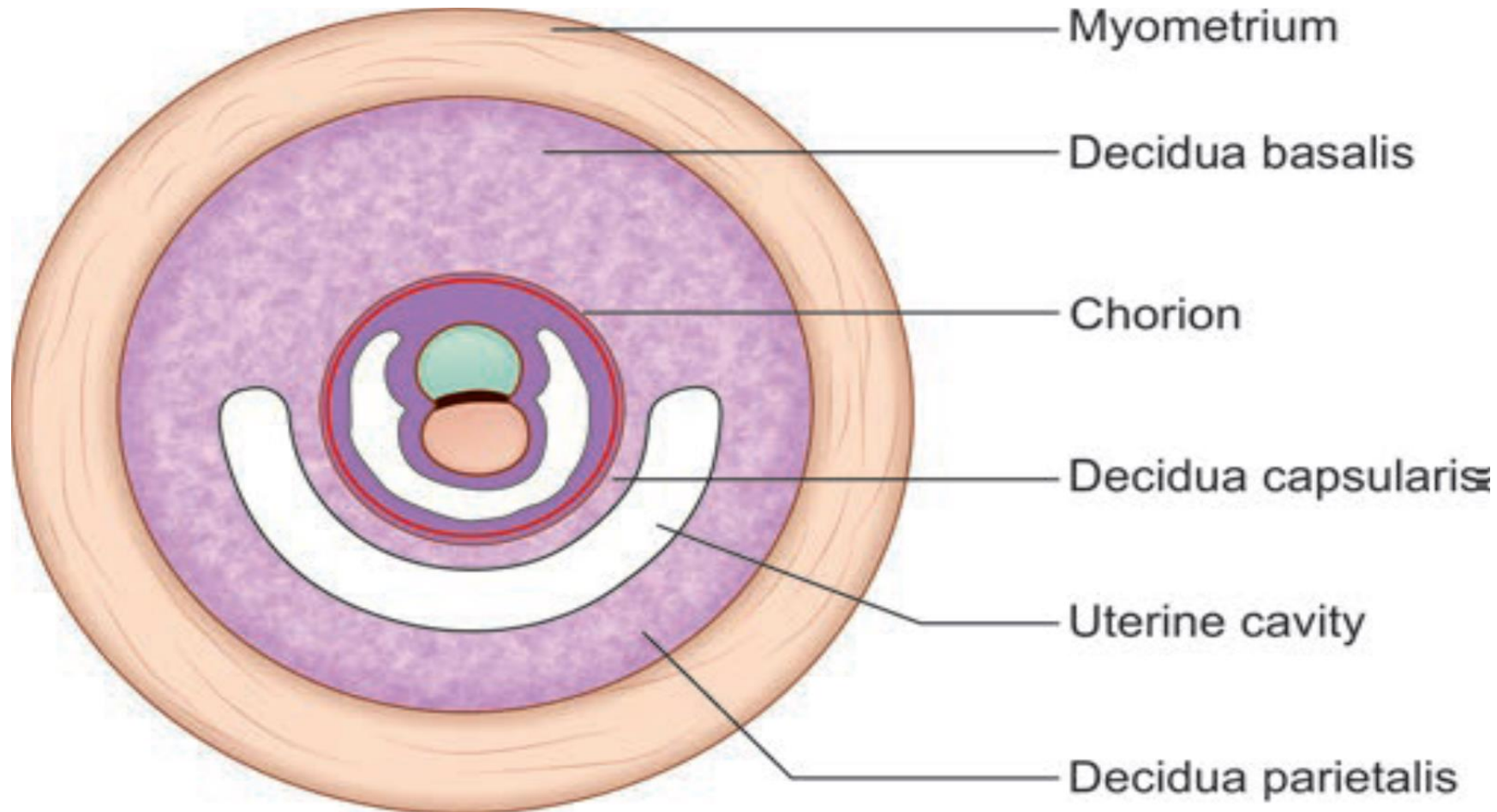
- **3 regions of the decidua** named according to their relation to the conceptus
- **1. *Decidua basalis/Serotina*:**
- part that contributes for the maternal component of placenta
- lies deep to the developing blastocyst
- Maternal blood vessels (spiral arteries) proliferate in the region of decidua basalis and are filled with blood and dilate to form sinusoids



- consists of large decidual cells that contain large amounts of lipids and glycogen (a source of nutrition for the embryo)
- also called *decidual plate* and firmly united to the chorion
- **2. *Decidua capsularis/Reflexa*:**
- surrounds the embryo like a capsule and separates it from the uterine cavity
- **3. *Decidua parietalis/Vera*:**
- decidua that lines the rest of uterine cavity.



- ***Fate of decidua:***
- As the conceptus enlarges during development, the decidua capsularis enlarges into the uterine cavity and finally fuses with decidua parietalis during 3rd month of pregnancy thus obliterating the uterine cavity



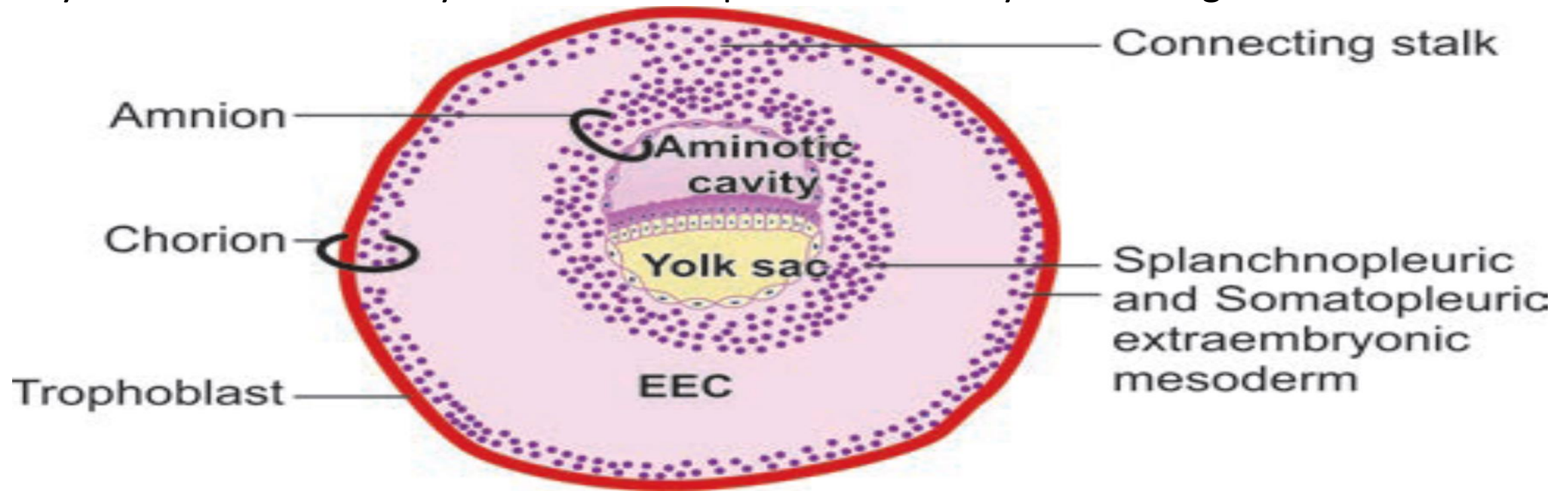
- **Chorion and Formation of Chorionic Villi**

- ***Chorion***

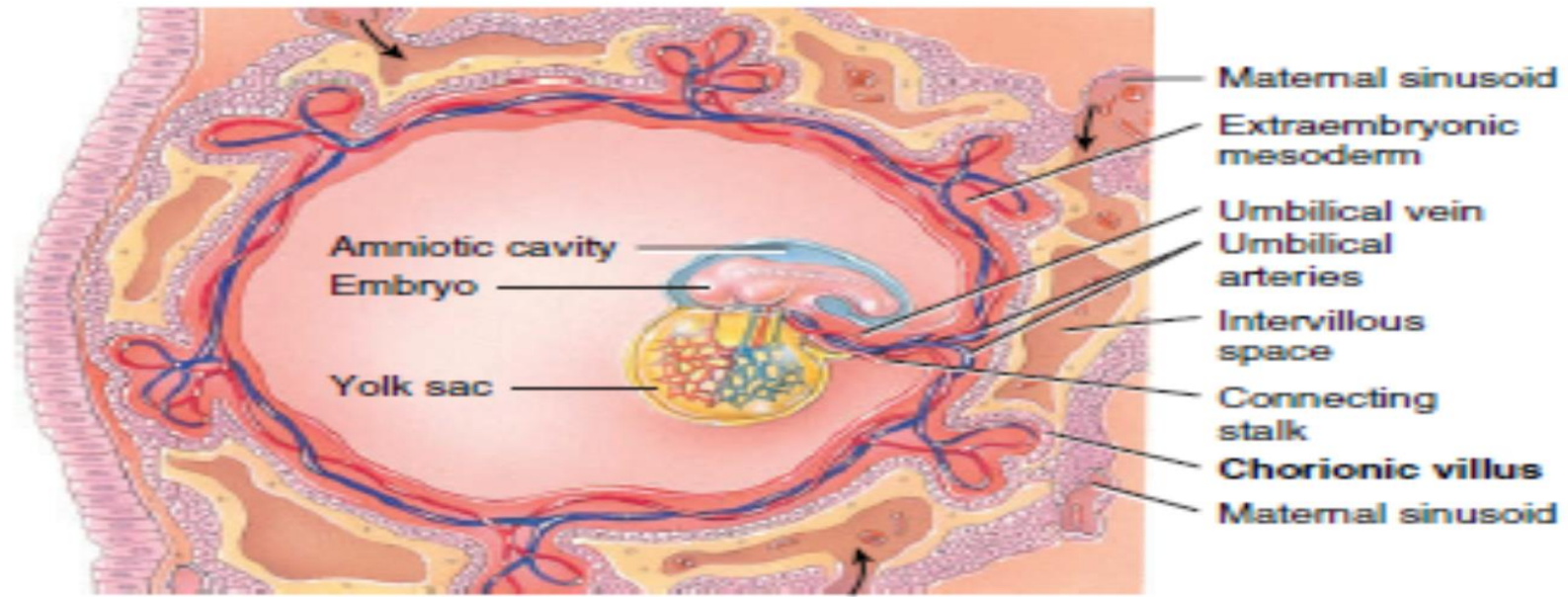
- cellular, outermost extraembryonic membrane composed of trophoblast lined with extraembryonic somatopleuric mesoderm

- ***Formation***

- extraembryonic somatic mesoderm & 2 layers of trophoblast (cytotrophoblast & syncytiotrophoblast) contribute for formation of chorion all around the developing embryo
- Embryo & its amniotic and yolk sacs are suspended into it by connecting stalk

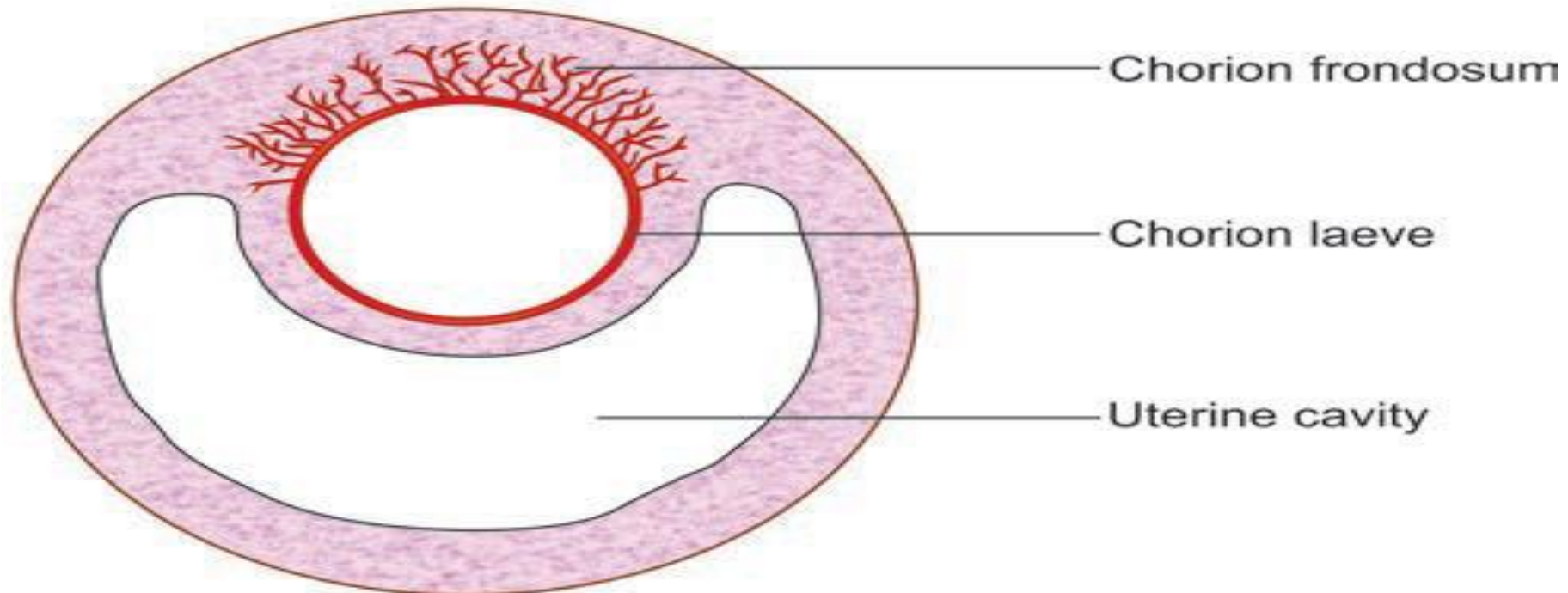
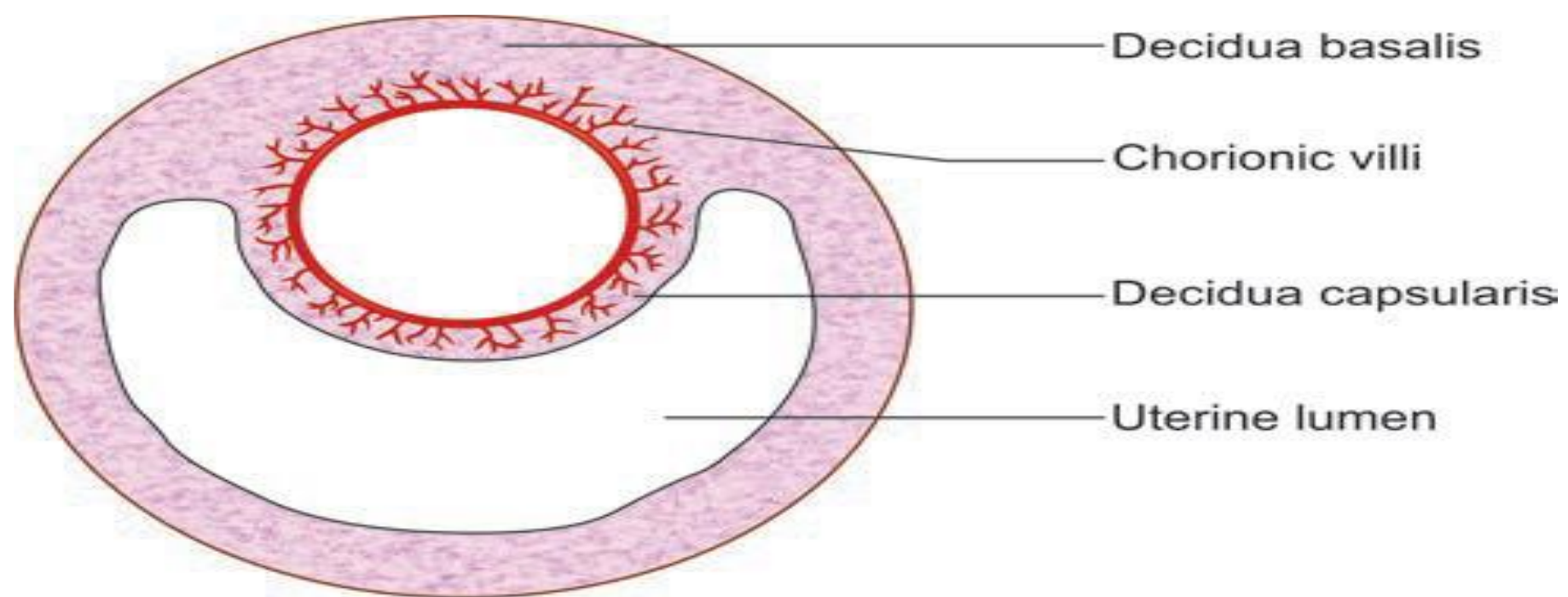


- amniotic sac with embryonic epiblast forms its floor & yolk sac with embryonic hypoblast forms its roof
- blood vessels develop in the extraembryonic mesoderm which forms the chorionic vessels
- **chorionic villi develop from the chorion** and cover the entire chorionic sac until the **beginning of 8th week** become vascularized by allantoic vessels

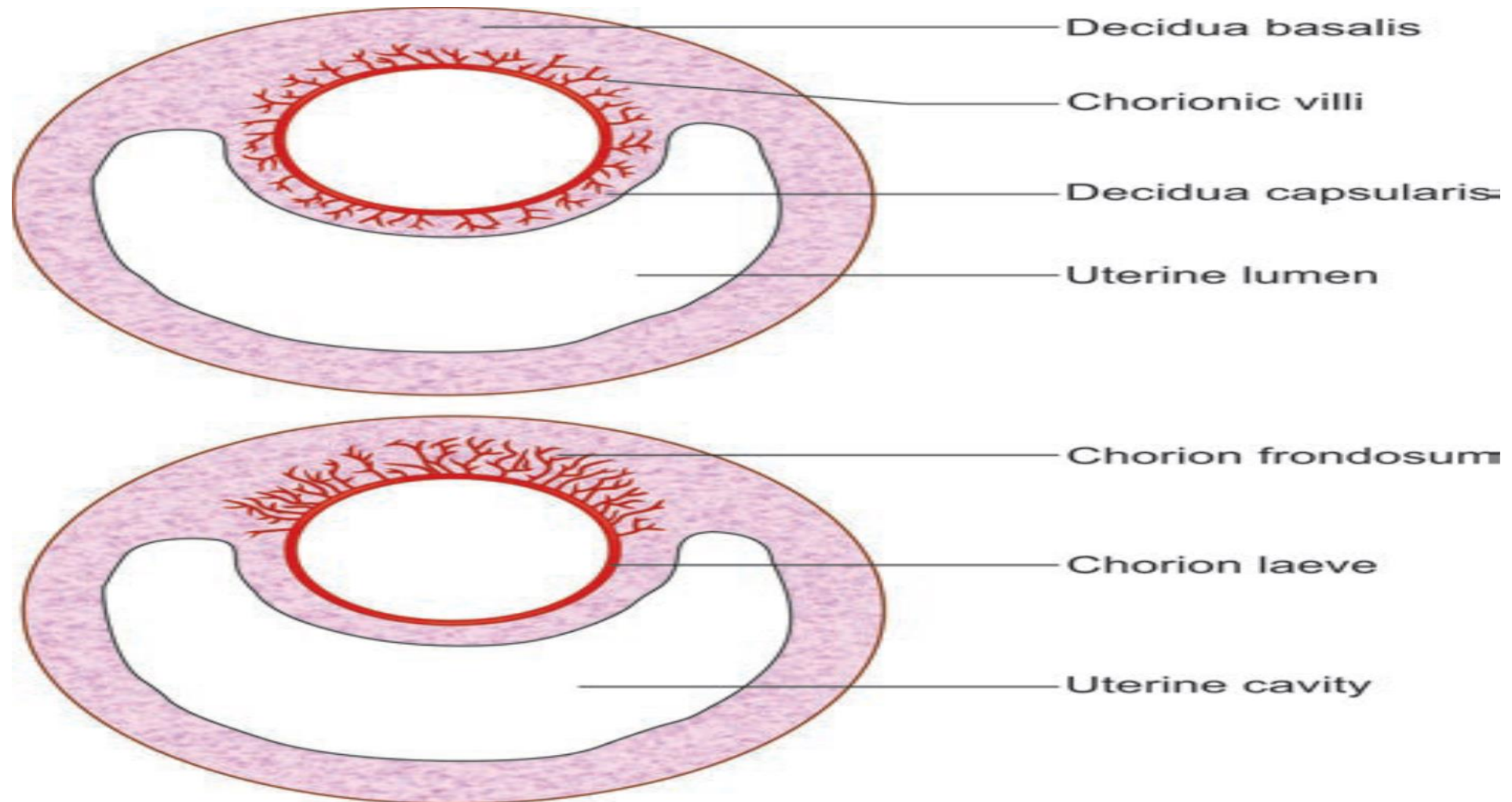


(c) Frontal section through uterus showing an embryo and its vascular supply, about 21 days after fertilization

- **Types of chorion**
- chorionic villi are first formed all over the trophoblast & grow into surrounding decidua
- **1. *Chorion laeve***: villi associated with decidua capsularis are transitory
- As the gestational sac grows they get compressed & their blood supply is reduced
- reduced blood supply these villi degenerate producing an avascular bare area of smooth chorion
- chorion laeve regresses in 3rd month of pregnancy

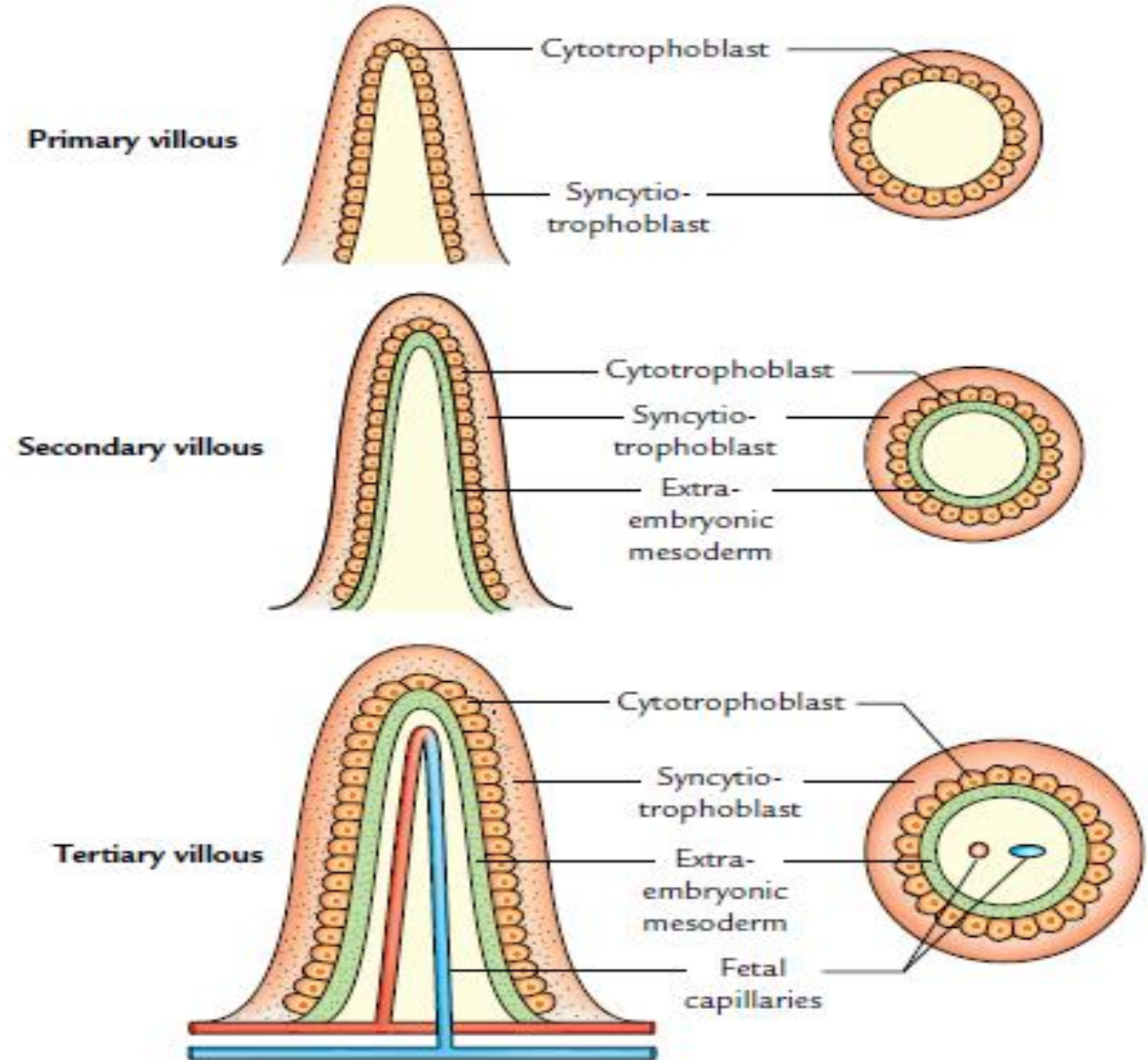
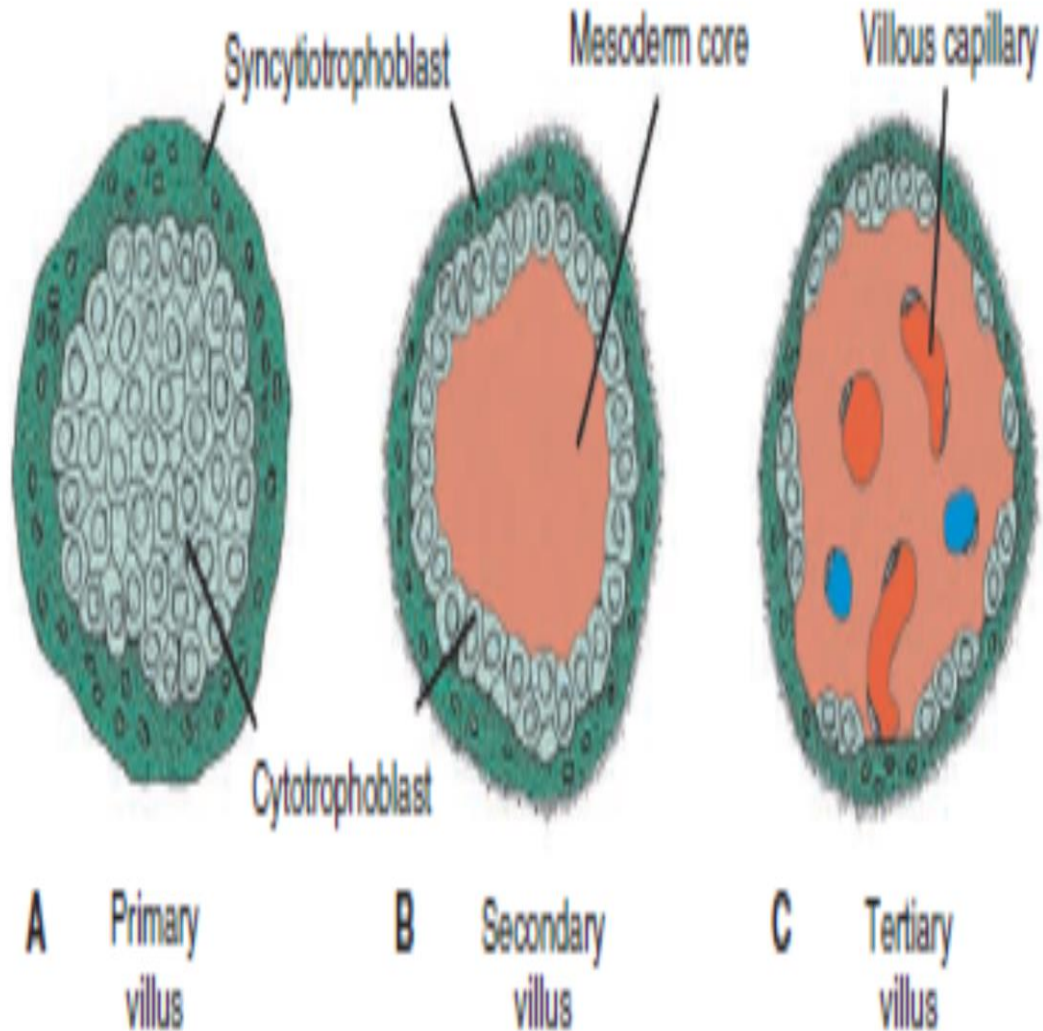


- ***Chorion frondosum***: chorionic villi associated with decidua basalis retain the vascularity, develop and form an area called *chorion frondosum*
- contributes fetal part of placenta as the maternal part is contributed by decidua basalis

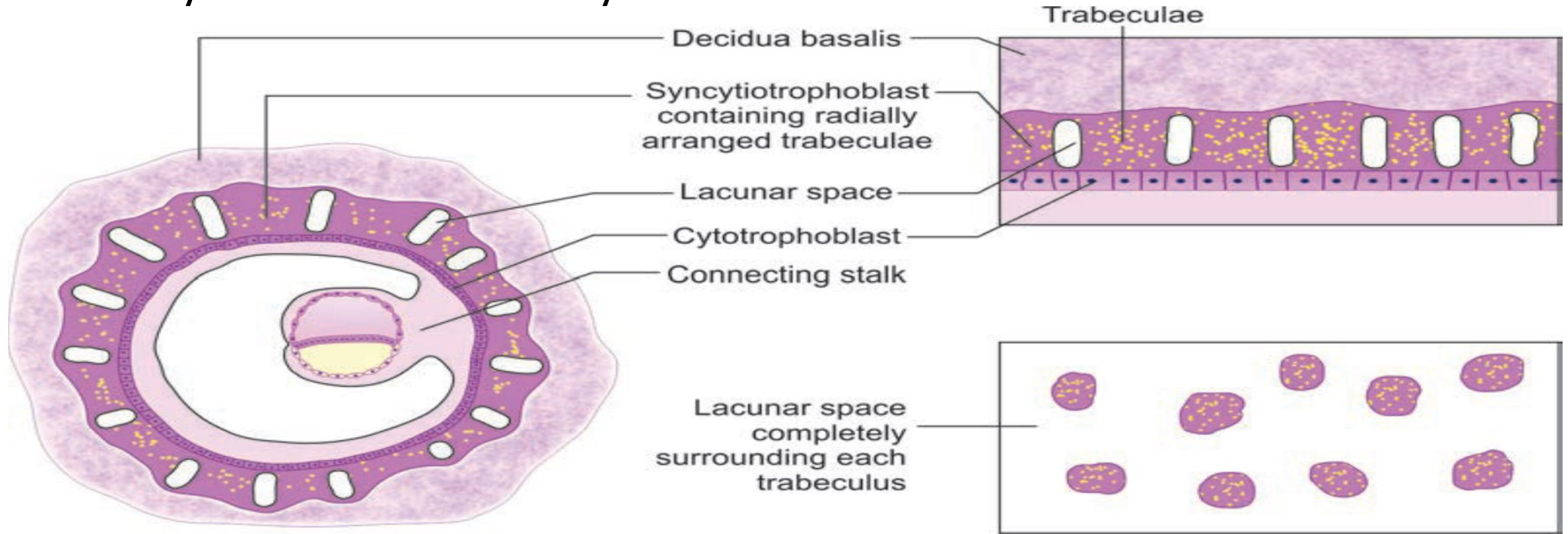


- ***Stages in the Formation of Chorionic Villi***
- structural component of chorionic villus differs at different periods of embryonic development
- 3 types of chorionic villi
- 1. *Primary villi:*
- consist of a central core of cytotrophoblast covered by a layer of syncytiotrophoblast
- Adjoining villi are separated by an intervillous space
- 2. *Secondary villi:*
- has 3 layers. Outer syncytiotrophoblast, an intermediate cytotrophoblast, and inner layer of somatopleuric extraembryonic mesoderm

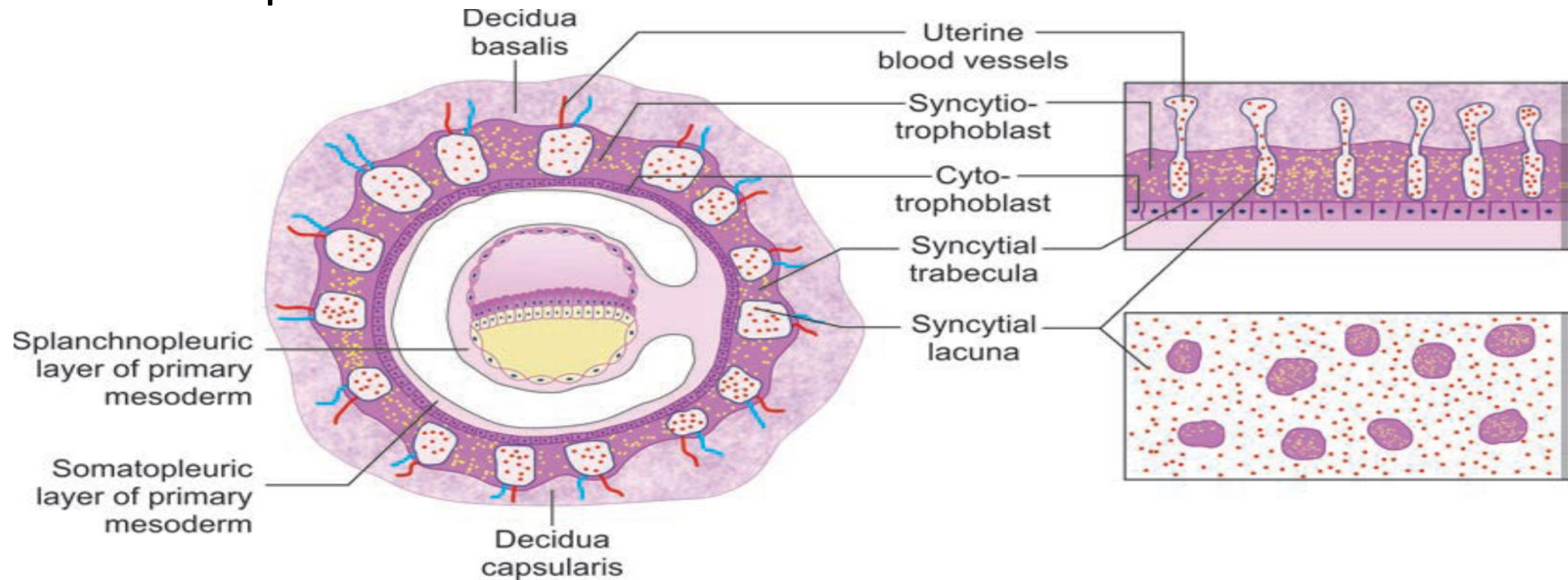
- 3. *Tertiary villi*:
- like secondary villi except that there are fetal blood capillaries in the mesoderm



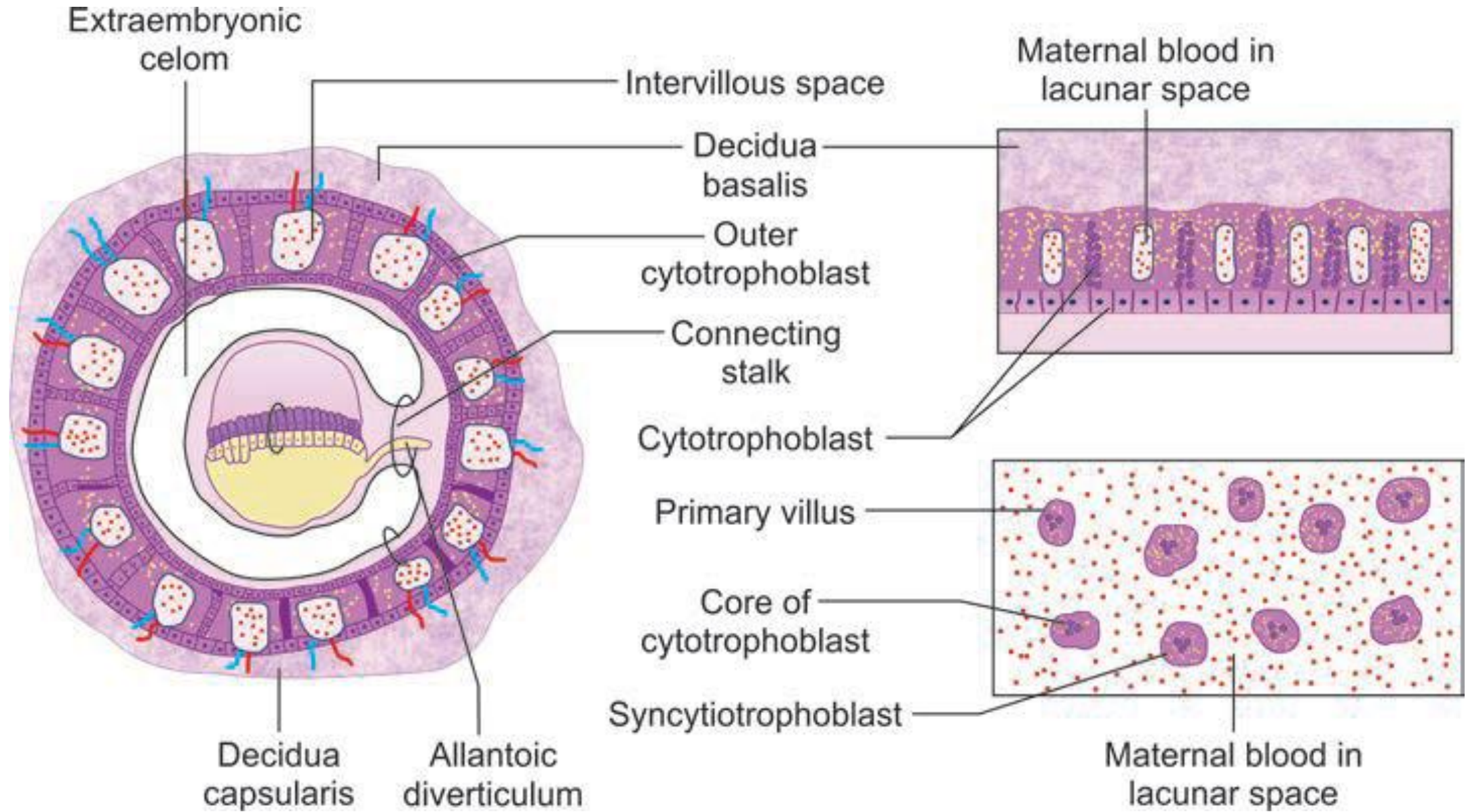
- ***Process of villus formation:***
- ***Appearance of lacunae and trabeculae in syncytiotrophoblast:***
- syncytiotrophoblast grows and becomes thick
- Small cavities called lacunae appear in this layer
- Lacunae increase in size, they are irregularly arranged
- lie radially around the blastocyst



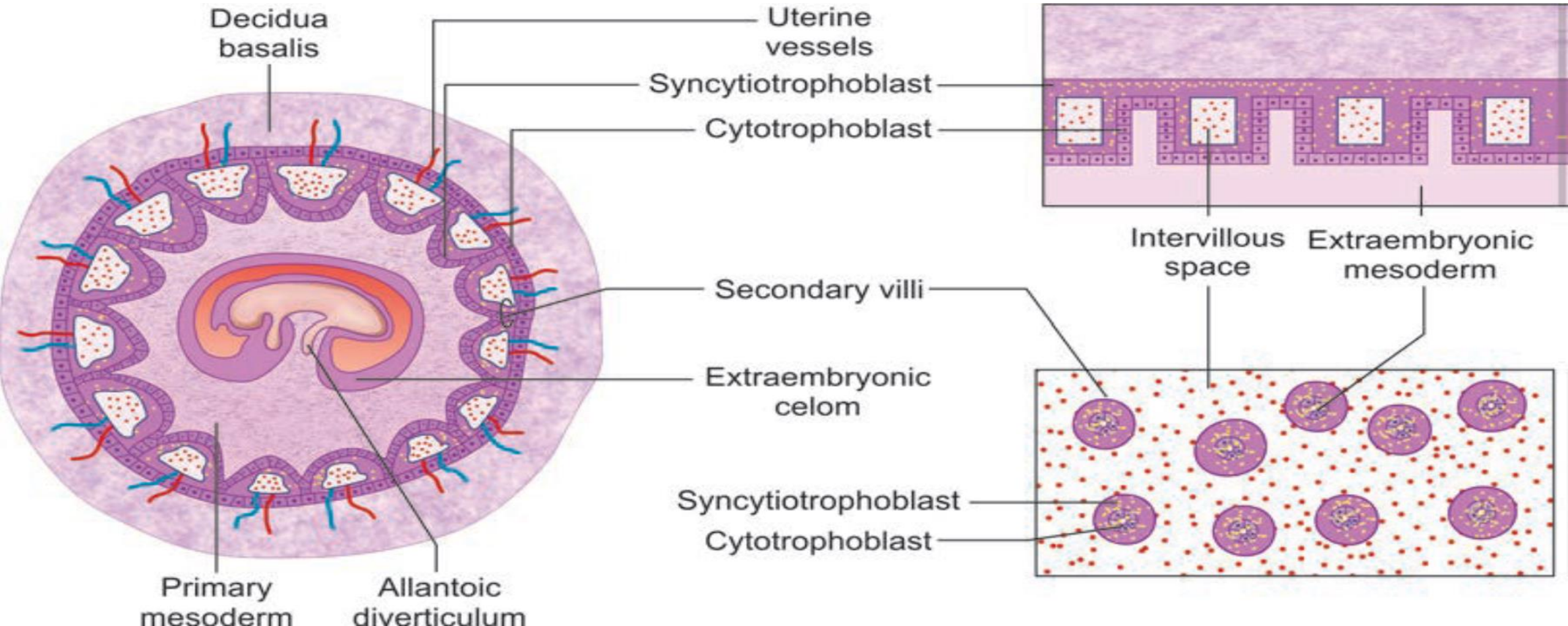
- lacunae are separated from one another by partitions of syncytium called **trabeculae**
- lacunae communicate with each other, one large space is formed
- Each trabeculus is surrounded all around by this lacunar space
- syncytiotrophoblast in which these changes occur grows into the endometrium
- endometrium is eroded, some blood vessels are opened up & blood from them fills the lacunar space



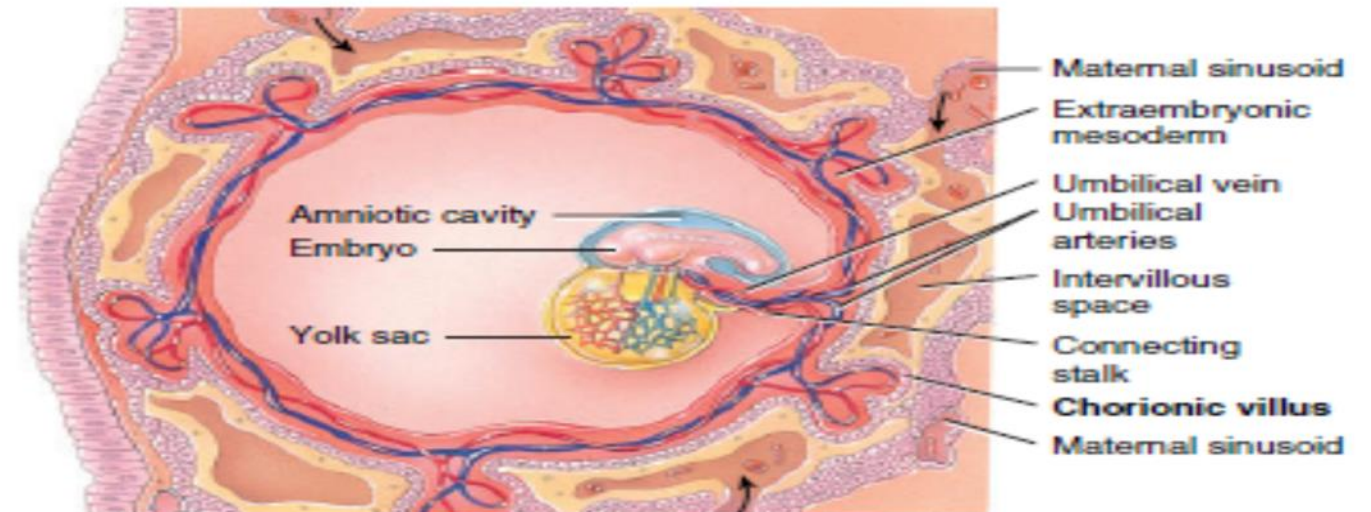
- ***Formation of primary chorionic villi***
- Each trabeculus is made up of Syncytiotrophoblast & surrounded by lacunar space filled with maternal blood
- cells of the cytotrophoblast multiply and grow into each trabeculus
- trabeculus now has a central core of cytotrophoblast covered by an outer layer of syncytium
- surrounded by maternal blood, filling the lacunar space
- trabeculus is called a ***primary villus*** and the lacunar space is now called the ***intervillous space***



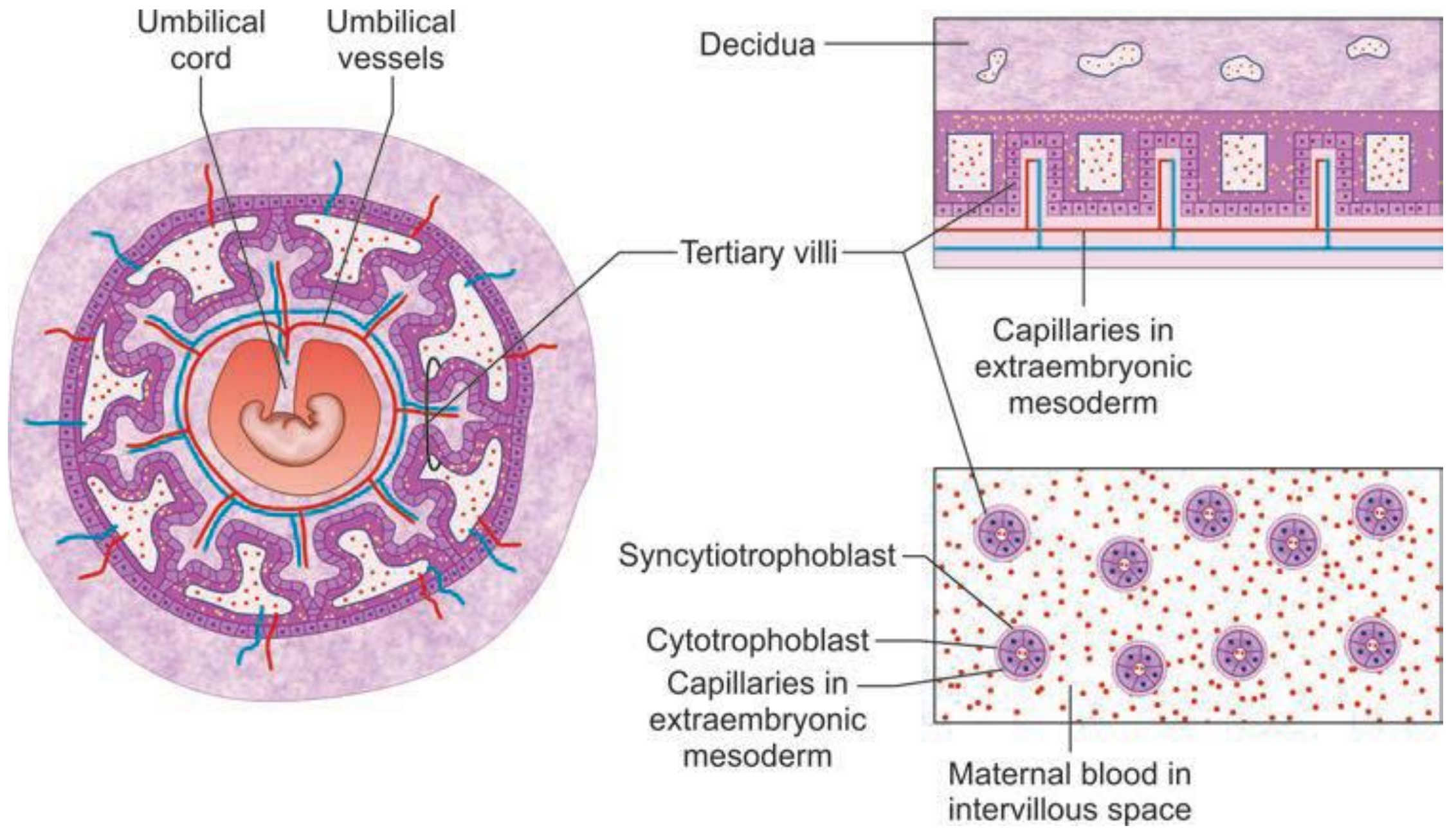
- ***Formation of secondary chorionic villi***
- Extraembryonic mesoderm invades the center of each primary villus
- villus has a core of mesoderm covered by cytotrophoblast and syncytium
- structure is now *secondary villus*



- ***Formation of tertiary chorionic villi :***
- blood vessels develop in mesoderm forming the core of each villus
- villus is fully formed & is now a *tertiary villus*
- ***Establishment of fetal blood circulation in tertiary villi:***
- blood vessels of the villus establish connections with the circulatory system of the embryo
- Fetal blood circulates through the villi, while maternal blood circulates through the intervillous space

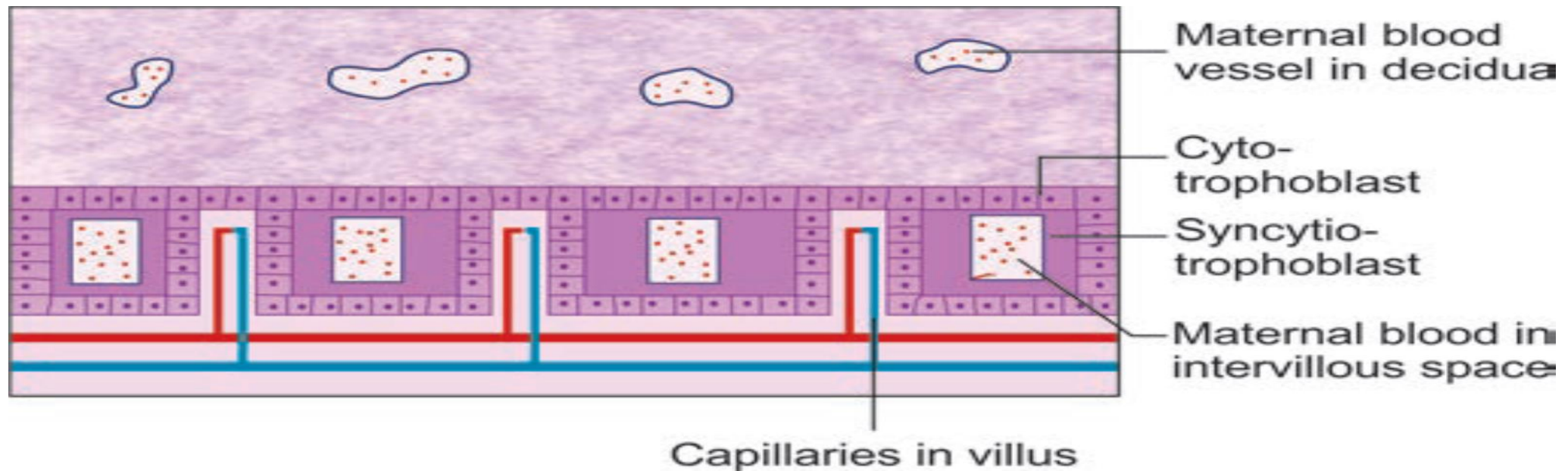


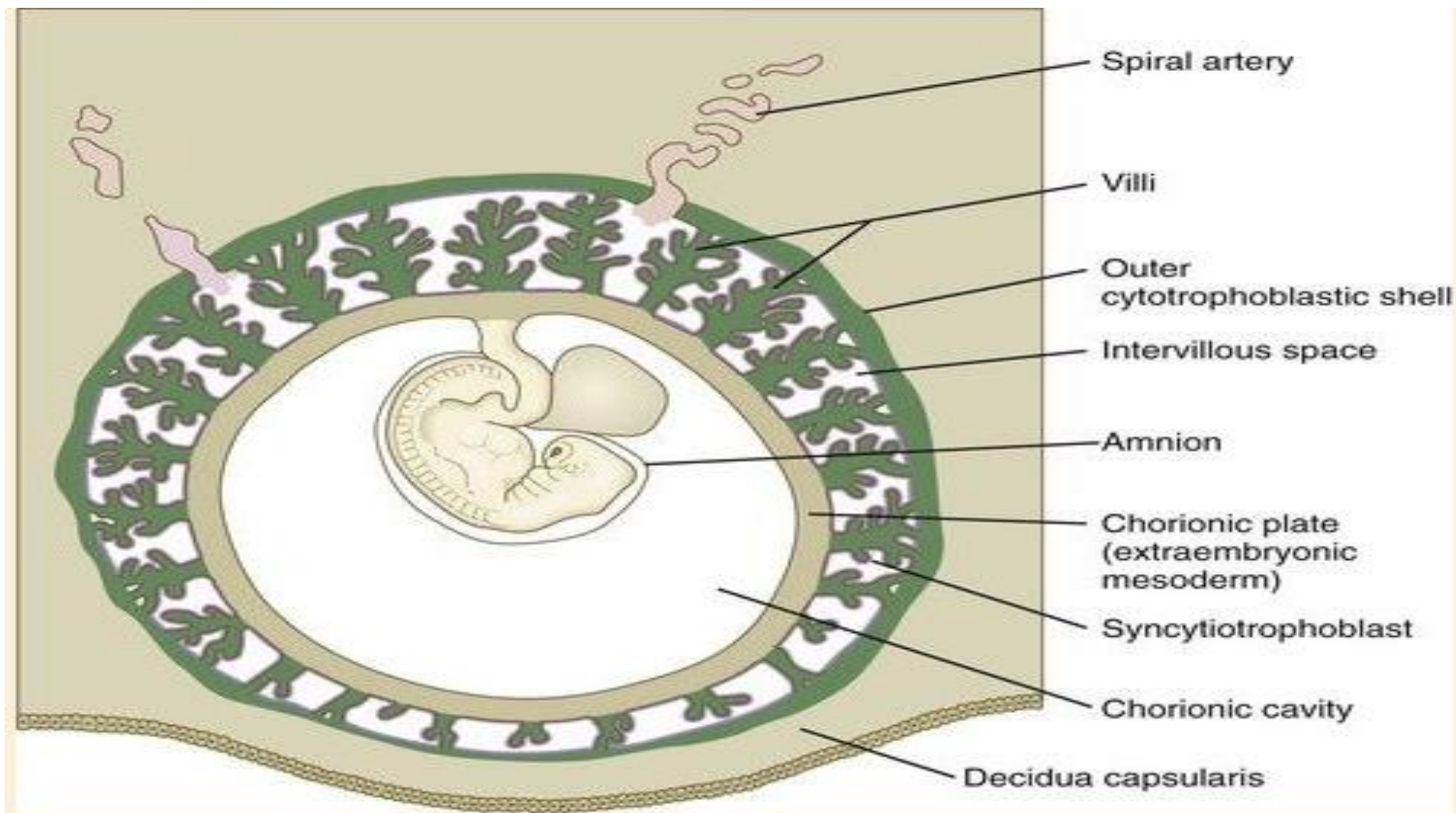
(c) Frontal section through uterus showing an embryo and its vascular supply, about 21 days after fertilization



- **Formation of cytotrophoblastic shell:**

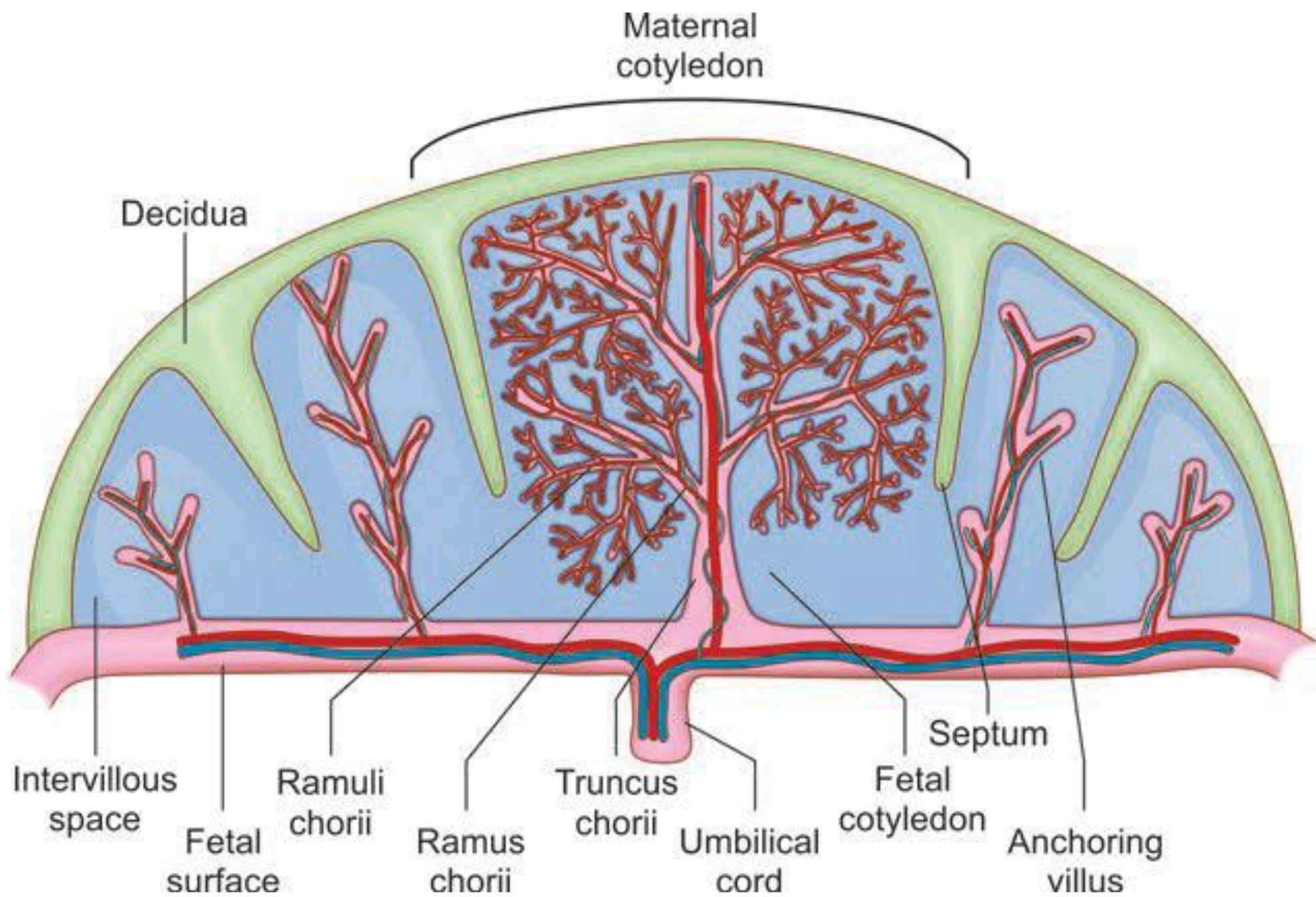
- cytotrophoblast grows into the villus does not penetrate the entire thickness of syncytium
- cytotrophoblast emerges through the syncytium of each villus
- cells of the cytotrophoblast now spread out to form a layer that completely cuts off the syncytium from the decidua
- layer of cells is called ***cytotrophoblastic shell***
- cells of this shell multiply and the placenta increases in size



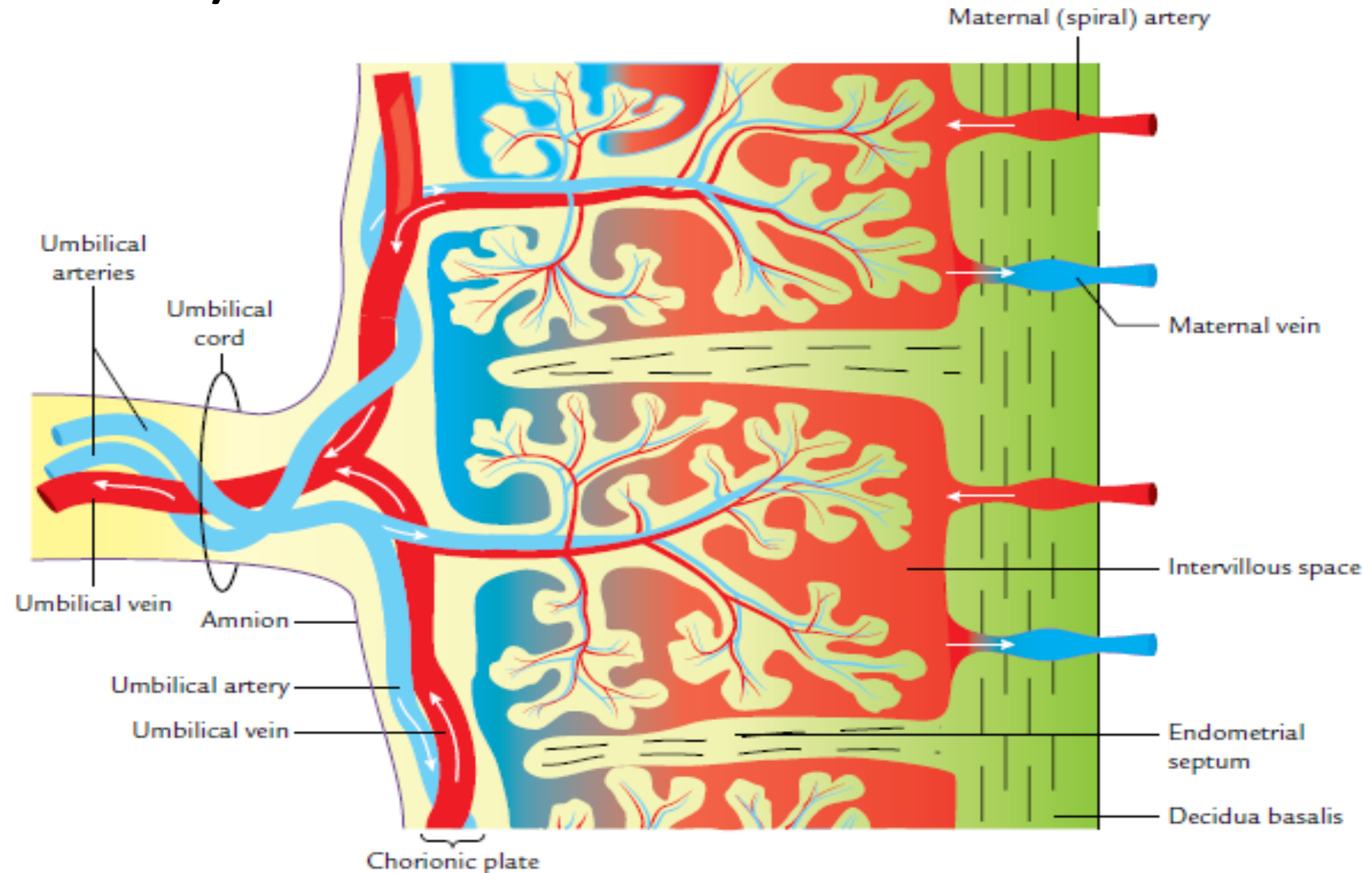


- ***Subdivisions of Villus***

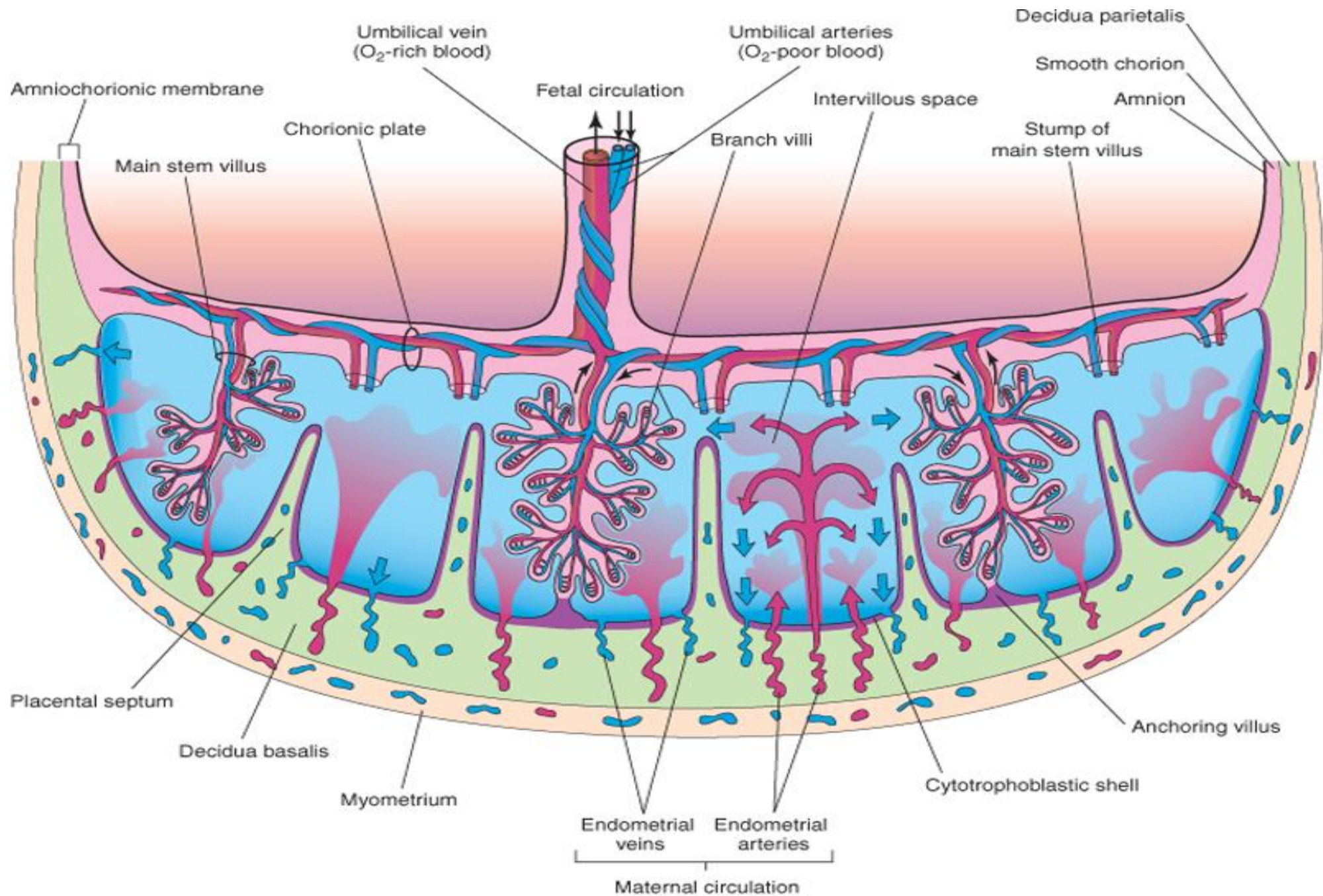
- villi first formed are attached on the fetal side to the **extraembryonic mesoderm** and on the maternal side to the **cytotrophoblastic shell** are called ***anchoring villi***
- Each anchoring villus consists of a *stem villus* or ***truncus chorii***
- Each stem villus divides into a number of branches called ***rami chorii***
- rami chorii in turn divide into finer branches called ***ramuli chorii***
- ramuli *chorii* are attached to the cytotrophoblastic shell
- anchoring villi give off numerous branches that grow into the intervillous space as ***free/floating villi***
- Expression of the transcription factor Gcm1 (glial cells missing-1) in trophoblast stem cells regulates the branching process of the stem villi to form the vascular network in the placenta



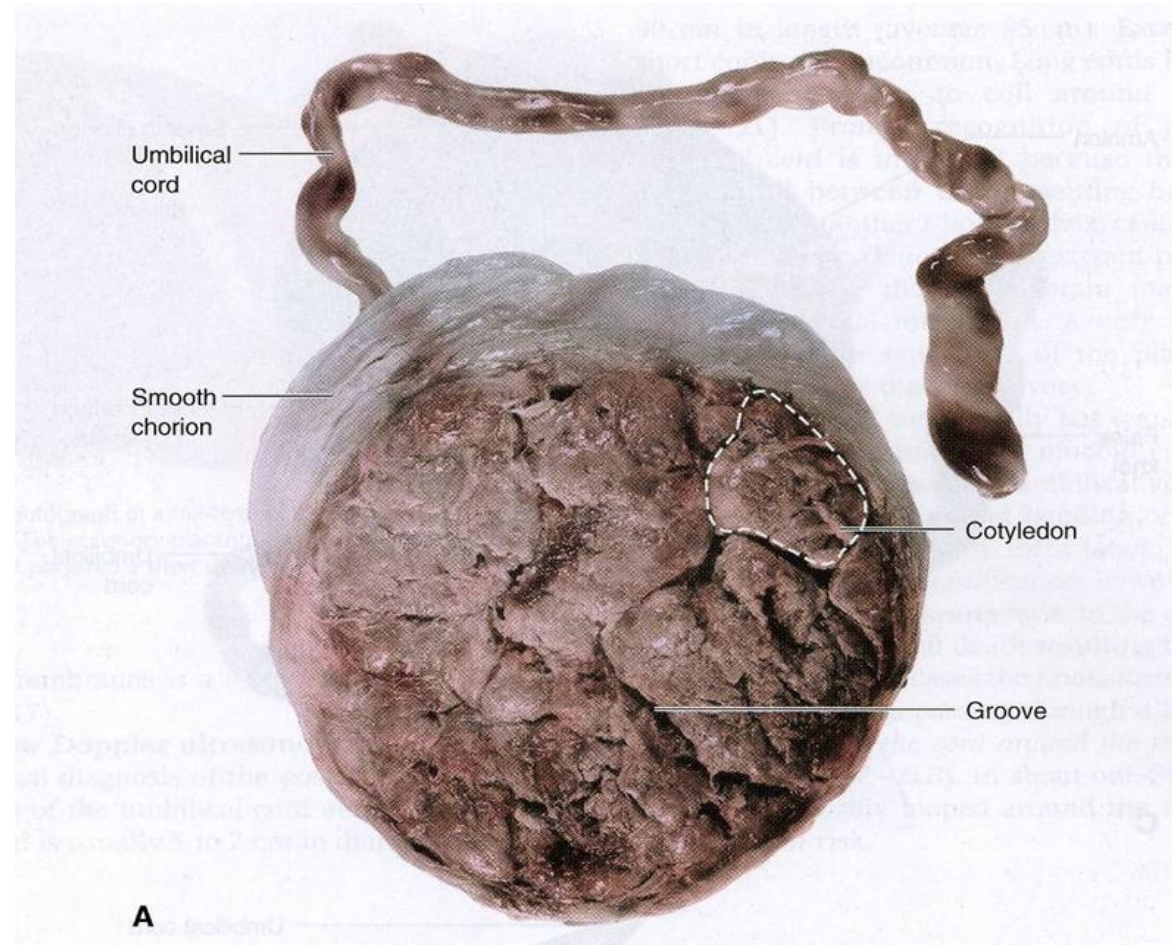
- The placenta presents:-
- **2 parts (fetal and maternal)**
- **2 surfaces (fetal and maternal)**
- **2 types of cotyledons**
- **a peripheral margin**



- ***Maternal part:*** contributed by decidua basalis or decidual plate of endometrium
- The maternal surface is rough and irregular
- subdivided into a number of lobes called ***maternal cotyledons***
- Septa that grow into the intervillous space from the maternal side divide this surface into 15–20 rough and irregular maternal cotyledons
- After birth, the placenta is always inspected for **missing cotyledons**. Cotyledons remaining attached to the uterine wall after birth may cause severe bleeding

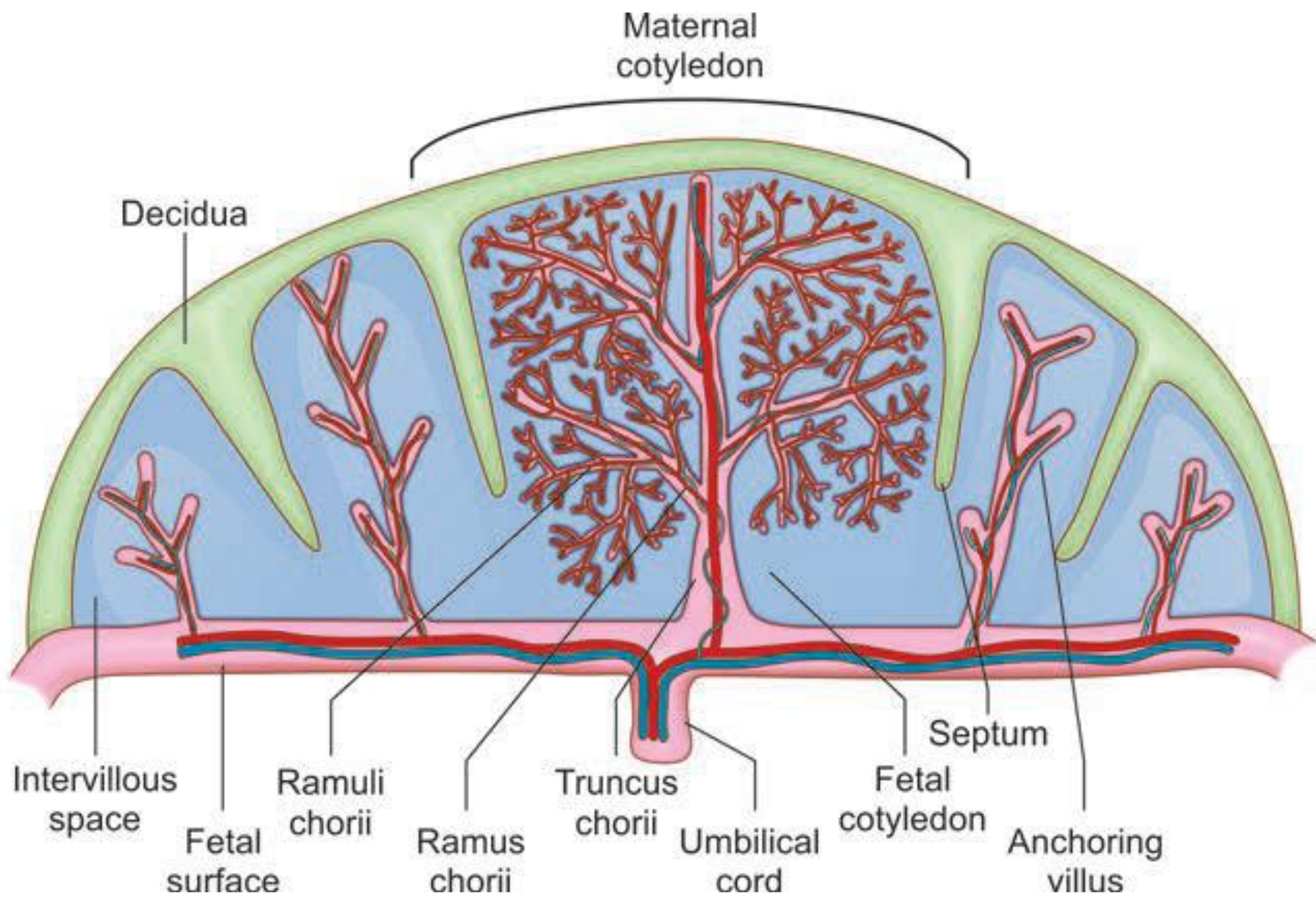


- the placenta viewed from the maternal side, the bases of the septa are seen as grooves while the cotyledons appear as convex areas bounded by the grooves

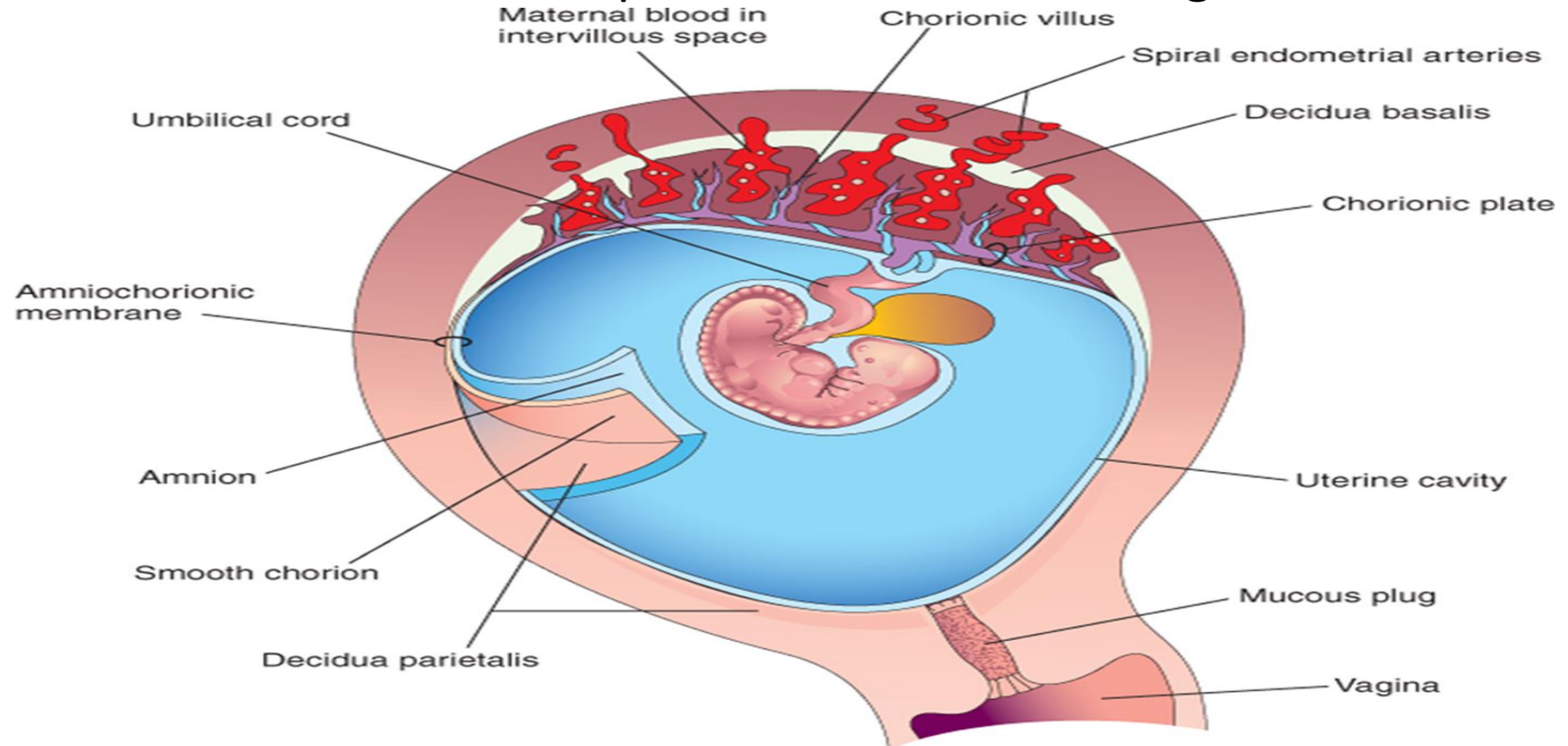


- ***Fetal surface:***
- surface is smooth and is covered by amnion
- umbilical cord is attached close to the center of this surface
- fetal part is contributed by chorionic frondosum seen as a plate called *chorionic plate*
- From the chorionic plate 40–60 extensions (fetal cotyledons) arise and extend toward the decidua basalis
- ***fetal cotyledon*** consists of a stem villus/truncus chorii that show ramifications into number of branches (ramus chorii), each further subdivides (ramuli chorii) like the branches of a tree

- Terminal ramifications look like fingers and are called chorionic villi
- villi that are attached to decidua basalis are called *anchoring villi*
- Others float in the maternal blood that flows in between the villi and are called *floating villi*
- ***Maternal and fetal cotyledons:***
- 15–20 maternal cotyledons in placenta
- Each maternal cotyledon contains 2–4 anchoring villi and their branches
- 1 anchoring villus and its ramifications (ramus chorii, ramuli chorii and floating villi) constitute a fetal cotyledon



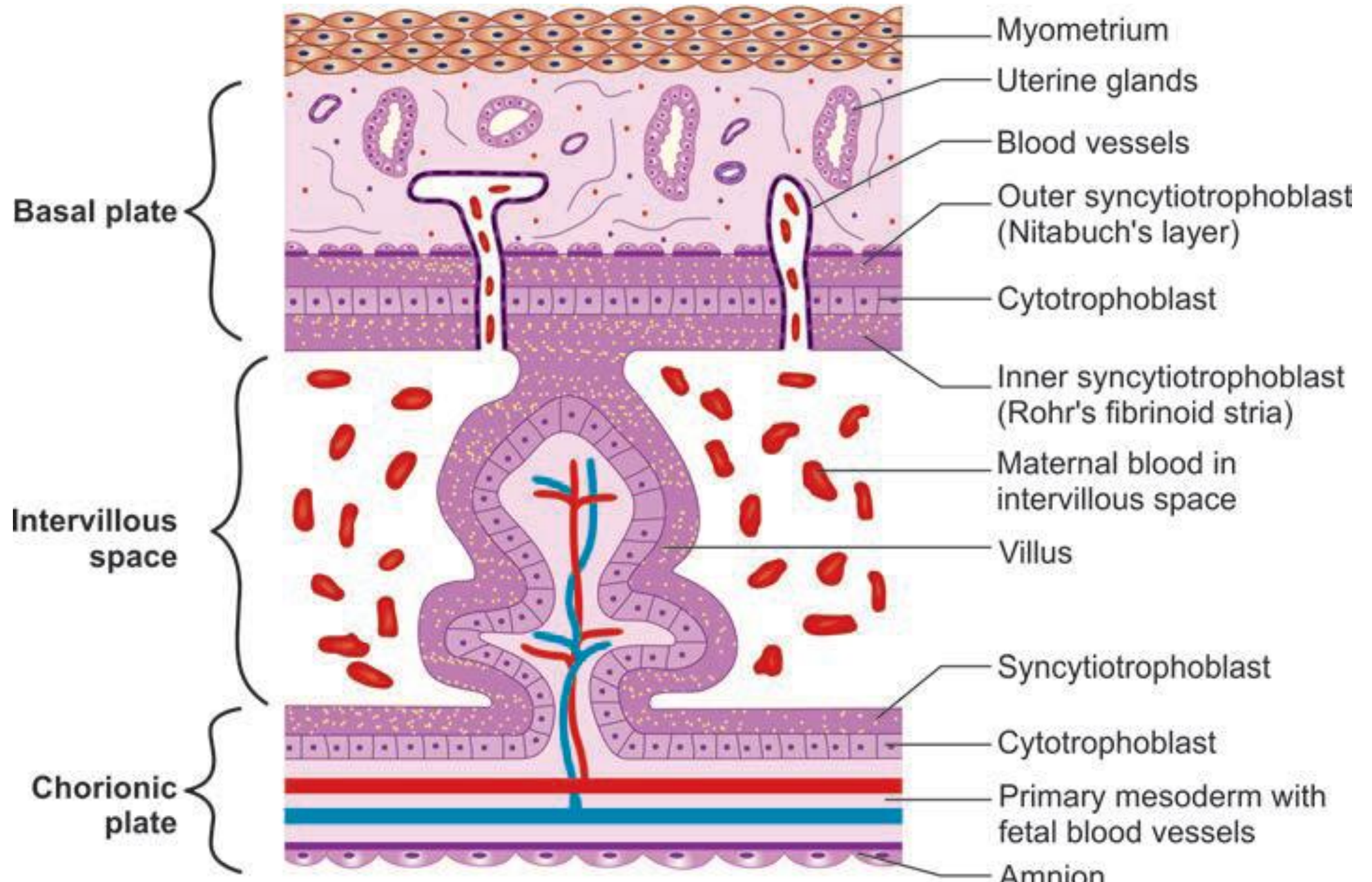
- ***Peripheral margin of placenta:***
- presents fetal membrane which is contributed from inside outward by decidua capsularis and parietalis, chorion laeve and amnion
- After the birth of the child, the placenta is shed off along with the decidua



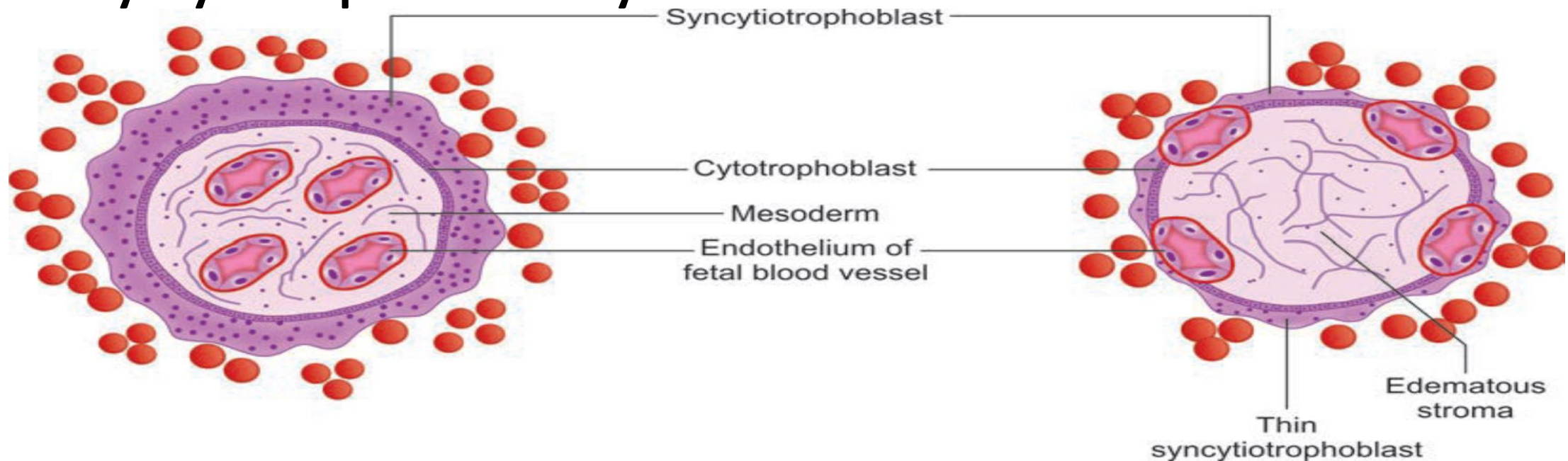
- **Placental Membrane/Barrier**

- maternal blood circulates through the intervillous space and fetal blood circulates through blood vessels in the villi
- maternal and fetal bloods are flowing side by side and in opposite directions they do not mix with each other
- separated by a membrane, made up of the layers of the wall of the villus
- Tissues intervening between fetal blood in chorionic villi and maternal blood in intervillous space constitute the placental membrane or barrier
- All interchanges of oxygen, nutrition and waste products take place through this membrane

- constituent structures forming the placental barrier or maternal fetal barrier extending from the maternal erythrocyte to fetal erythrocyte are as follows:
- In the early part of pregnancy, the barrier presents the following layers
- **Endothelium of fetal blood vessels, and its basement membrane**
- **Surrounding mesoderm (connective tissue)**
- **Cytotrophoblast and its basement membrane**
- **Syncytiotrophoblast**



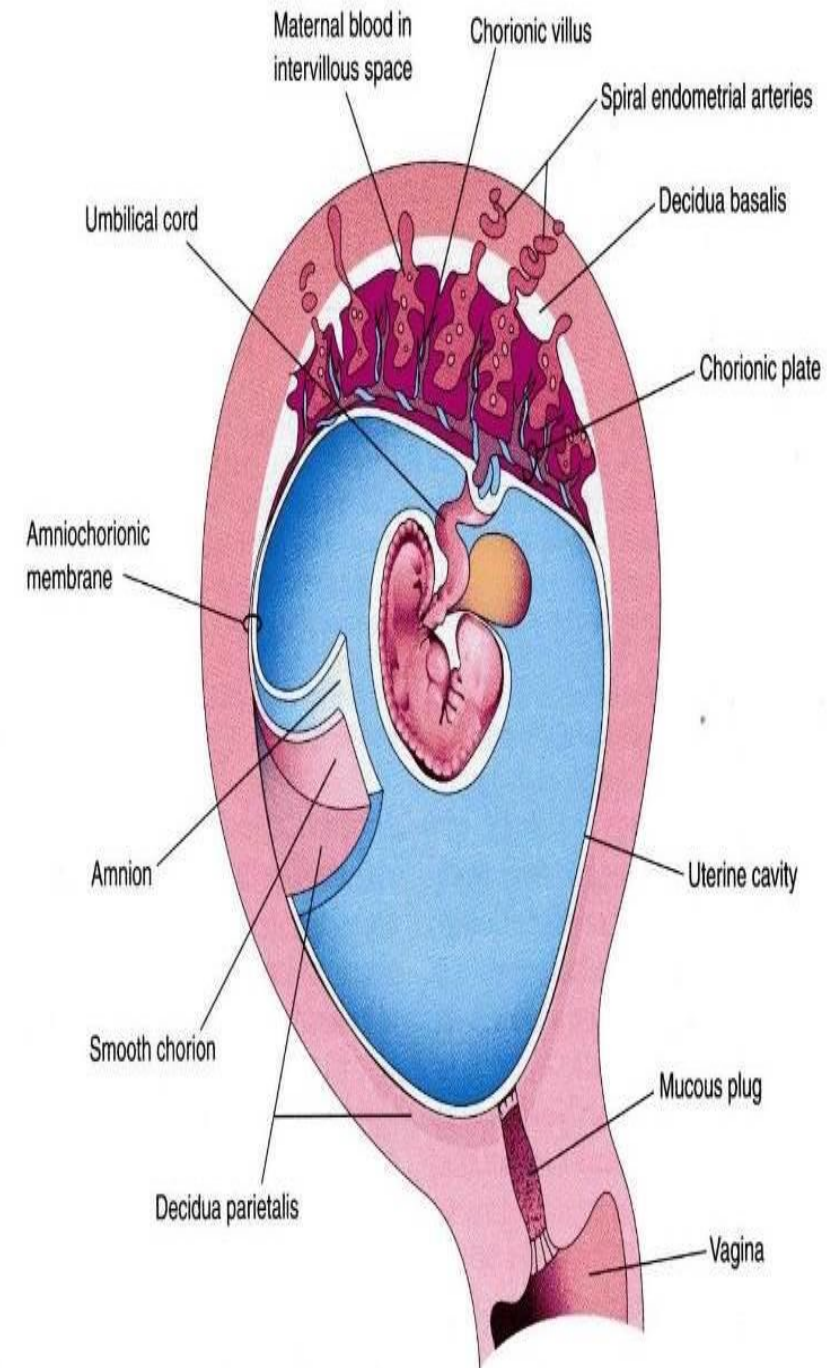
- later part of pregnancy, efficiency of the membrane is increased due to the reduction in its thickness by
- **Disappearance of cytotrophoblast**
- **Thinning of syncytiotrophoblast**
- **Edematous stroma**
- **Peripheral migration of fetal blood vessels**
- **It presents endothelium of fetal capillaries resting on basement membrane and syncytiotrophoblast only**



PLACENTA

Function Of The Placenta:

1. Protection
2. Nutrition
3. Respiration
4. Excretion
5. Hormone production



PLACENTAL ENDOCRINE SYNTHESIS

- The syncytiotrophoblast synthesizes protein & steroid hormones

The protein hormones

- **a. human chorionic gonadotropin** has an effect similar to luteinizing hormone (LH) of the pituitary gland
- **b. human chorionic somatotropin** has an anti-insulin effect on maternal blood causing increased plasma level of glucose and amino acids in maternal blood, and enhances utilization of glucose by the fetus
- **c. human chorionic thyrotropin –increases the rate of metabolism through out pregnancy**
- d. human chorionic corticotropin –stimulates the pituitary gland to release(adrenocorticotrophic hormone) ACTH**

The steroid hormones

- e. Progesterone to maintain pregnancy
- f. Estrogens which promotes uterine growth and development of mammary gland
- **6. Storage function:** That is placenta acts as a storage house for glycogen, calcium, and iron in early months of pregnancy.
- function is taken over by liver later

- **Classification of Placenta** -Based on shape

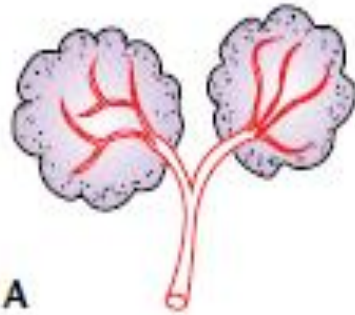
- Discoid—round or disc like
- Bidiscoidal—it consists of two discs
- — Oval
- — Triangular
- — Irregular
- – Lobed—it divides into lobes
- — Diffuse/placenta membranacea

chorionic villi persists all-round the blastocyst

- —Placenta succenturiata —a small part of the placenta is separated from the rest of it
- — Fenestrated —presence of hole or opening in the placenta
- — Circumvallate —when peripheral edge of placenta is covered by a circular fold of decidua, it is called circumvallate.

TYPES OF PLACENTA

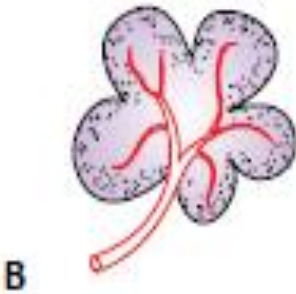
1. Bilobed/
Bidiscoidal



FEATURES

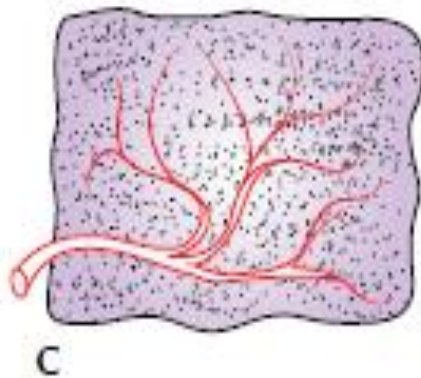
Two lobes

2. Multilobular



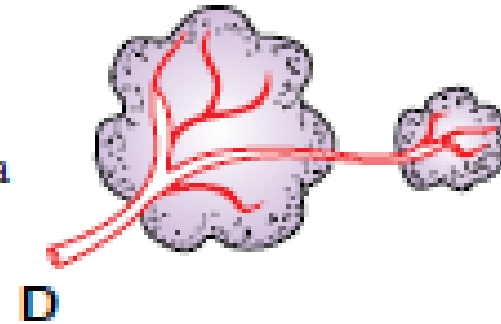
More than
two lobes

3. Diffuse



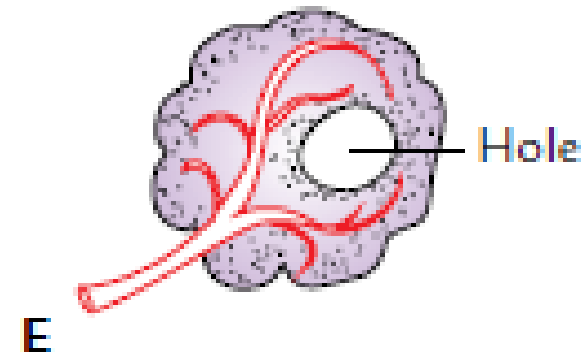
Thin and not
disc shaped

4. Placenta
succenturiata



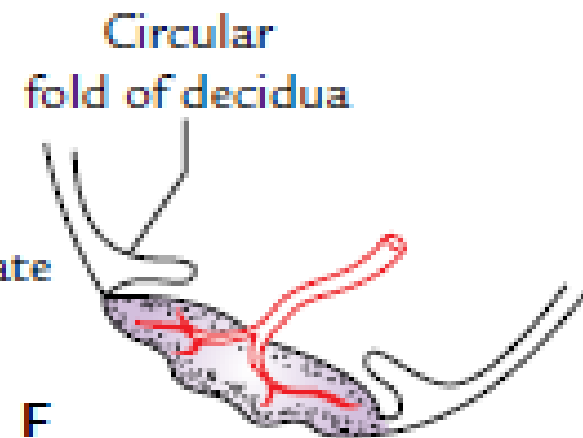
A small part of
placenta separated
from main placenta

5. Placenta
fenestrata



Presence of hole
in placental disc

6. Circumvallate
placenta

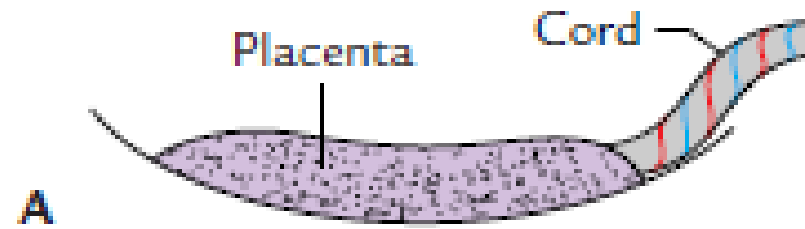


Peripheral edge of
placenta is covered
by circular fold of
decidua

ANOMALY OF PLACENTA

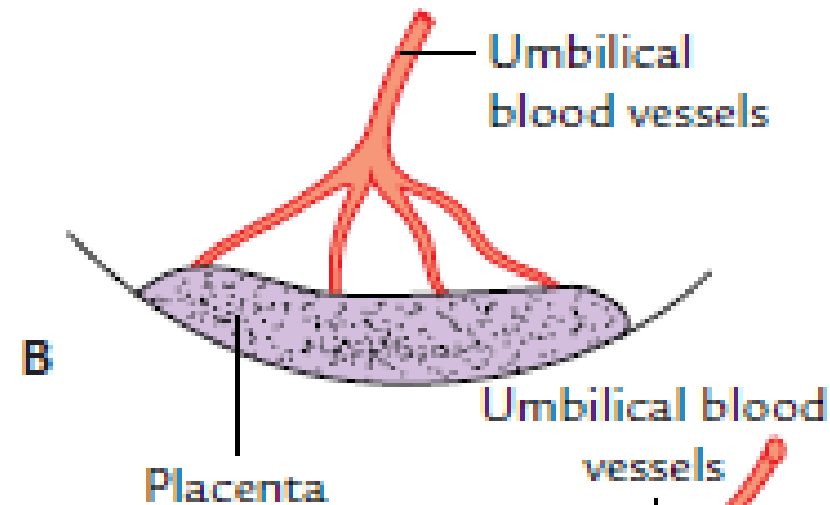
FEATURES

1. Marginal (battledore) placenta



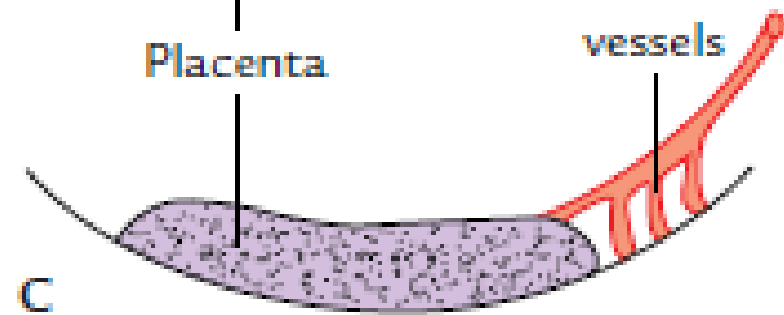
Umbilical cord attached to margin of placenta

2. Furcate placenta



Umbilical blood vessels divide before reaching placenta

3. Velamentous placenta



Umbilical blood vessels attached to amnion and ramify there before reaching placenta

Anomalies of placenta due to abnormal attachment of the umbilical cord.

- **FETAL/EXTRAEMBRYONIC MEMBRANES**

- Tissues or structures that develop from the zygote but do not form part of embryo proper
- Structures that constitute the fetal membranes are:
- **Trophoblast and chorion forming *placenta***
- ***Amnion*** or ectodermal vesicle covering the embryo or fetus and filled with amniotic fluid
- ***Yolk sac***—Primary, secondary, tertiary(definitive)
- ***Allantois*** or allantoenteric diverticulum
- **Connecting or body stalk and *umbilical cord***
- https://youtu.be/r_59vAmD05E

- Functions: These are concerned with protection, nutrition
- Protection—Amniotic membrane and fluid
- Respiration—Placenta
- Excretion
- Nutrition—Amniotic fluid, placenta, umbilical cord
- Clinical importance: essential for performing prenatal diagnostic procedures like chorionic villus biopsy and amniocentesis.

- **MULTIPLE BIRTHS AND TWINNING**

- *Multiple births*: more than one fetus is carried to term in a single pregnancy
- mother gives birth to 2 infants at the same time, are called twins.
- 3 (triplets), 4 (quadruplets) or even more infants are sometimes born simultaneously
- ***Types of twinning***:
 - Twins can be produced in 2 ways
 - ***Dizygotic twins***: 2 ova may be shed simultaneously from the ovary
 - Each of them may be fertilized and may develop in the usual manner.
 - twins are called *dizygotic* or *fraternal twins*

- each of them develops from a separate ovum, they have independent genetic constitutions
- need not be of the same sex, nor do they resemble each other
- Each fetus has its own chorionic and amniotic sacs
- Dizygotic twinning is more common in human beings than monozygotic twinning

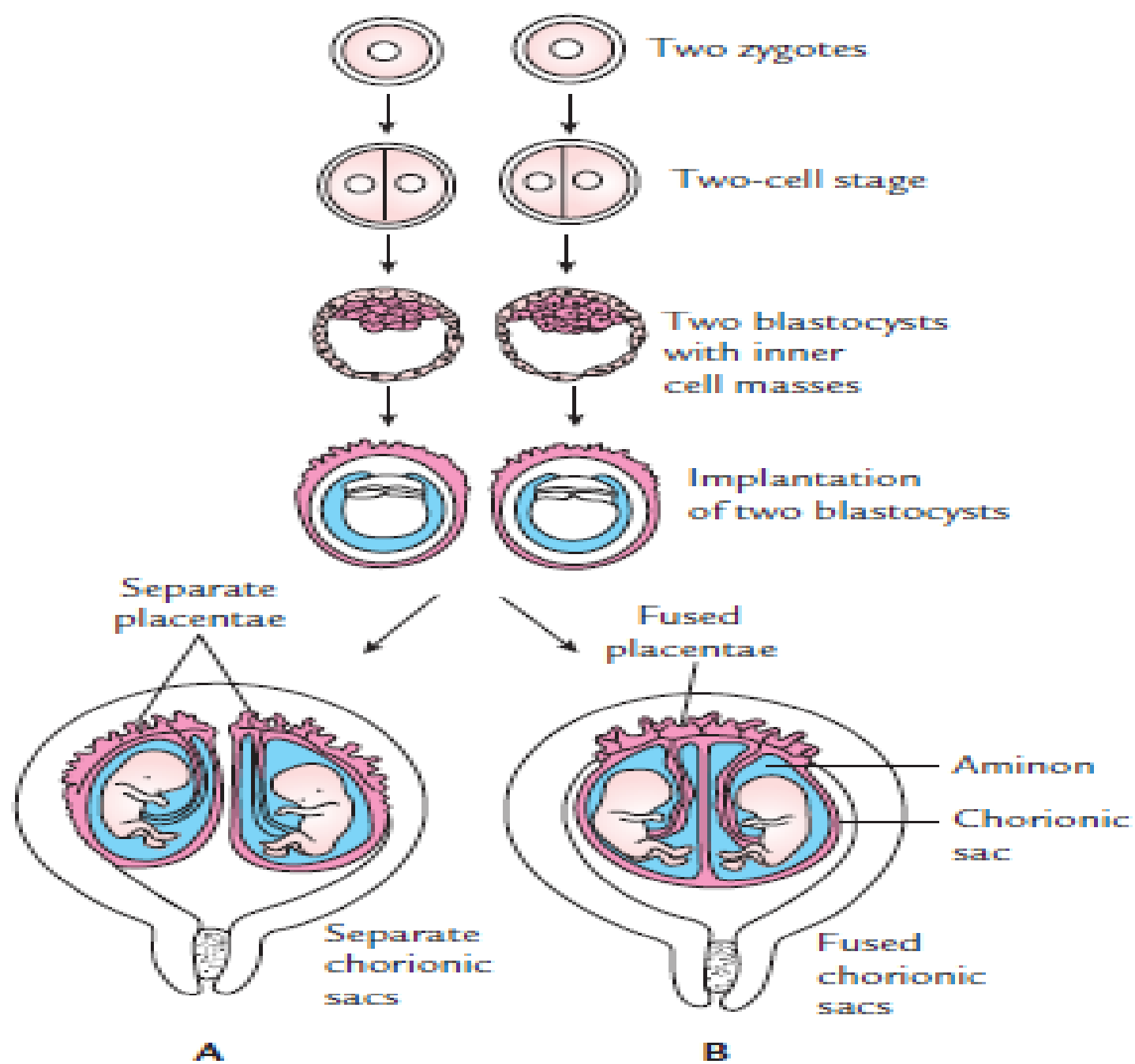


Fig. 6.22 Dizygotic twins.

- *Monozygotic twins*: arise from a single fertilized ovum
- genetic constitution of the two twins is exactly the same
- they are of the same sex.
- They are also exactly alike in appearance
- Monozygotic twins are produced in one of the following ways:
- — ***Early blastomere separation***:
- The cells formed in the first few divisions of the zygote are totipotent, each cell is capable of developing into a complete embryo
- two cells formed by the first division may separate and develop independently
- the fetuses will have separate chorionic and amniotic sacs
- incidence is 25–30%

- — ***Duplication of inner cell mass***: embryo may develop normally up to the stage of the morula
- when the blastocyst is formed, two inner cell masses form within it and each develops into a complete fetus
- the two fetuses have a common chorionic sac but each lies in an independent amniotic cavity
- percentage incidence is 70–75%

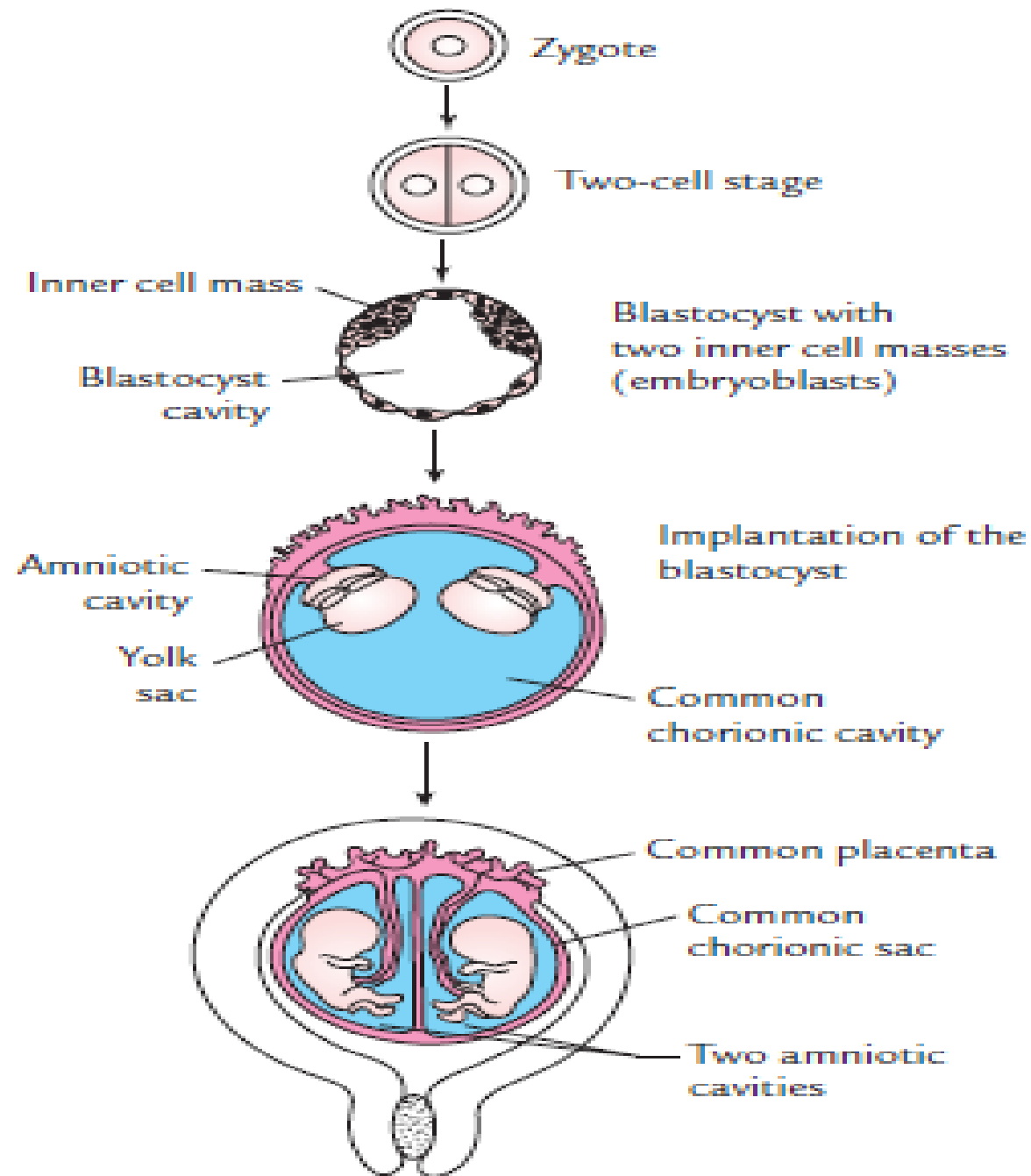


Fig. 6.23 Monozygotic twins.

- ***Duplication of embryonic disc***: The inner cell mass may split into two; or two embryonic axes may be established in one inner cell mass
- two separate embryonic discs are formed within it, each with its own prochordal plate and primitive streak
- the two fetuses share a common chorion as well as a common amniotic cavity
- incidence is 1–2%
- <https://youtu.be/GMRtjiZ9yo8>

- **Clinical correlation**
- ***Placenta previa***
- normal attachment of placenta is in the upper uterine segment
- attachment of placenta may extend partially or completely into the lower uterine segment
- due to the implantation of the blastocyst close to the internal os

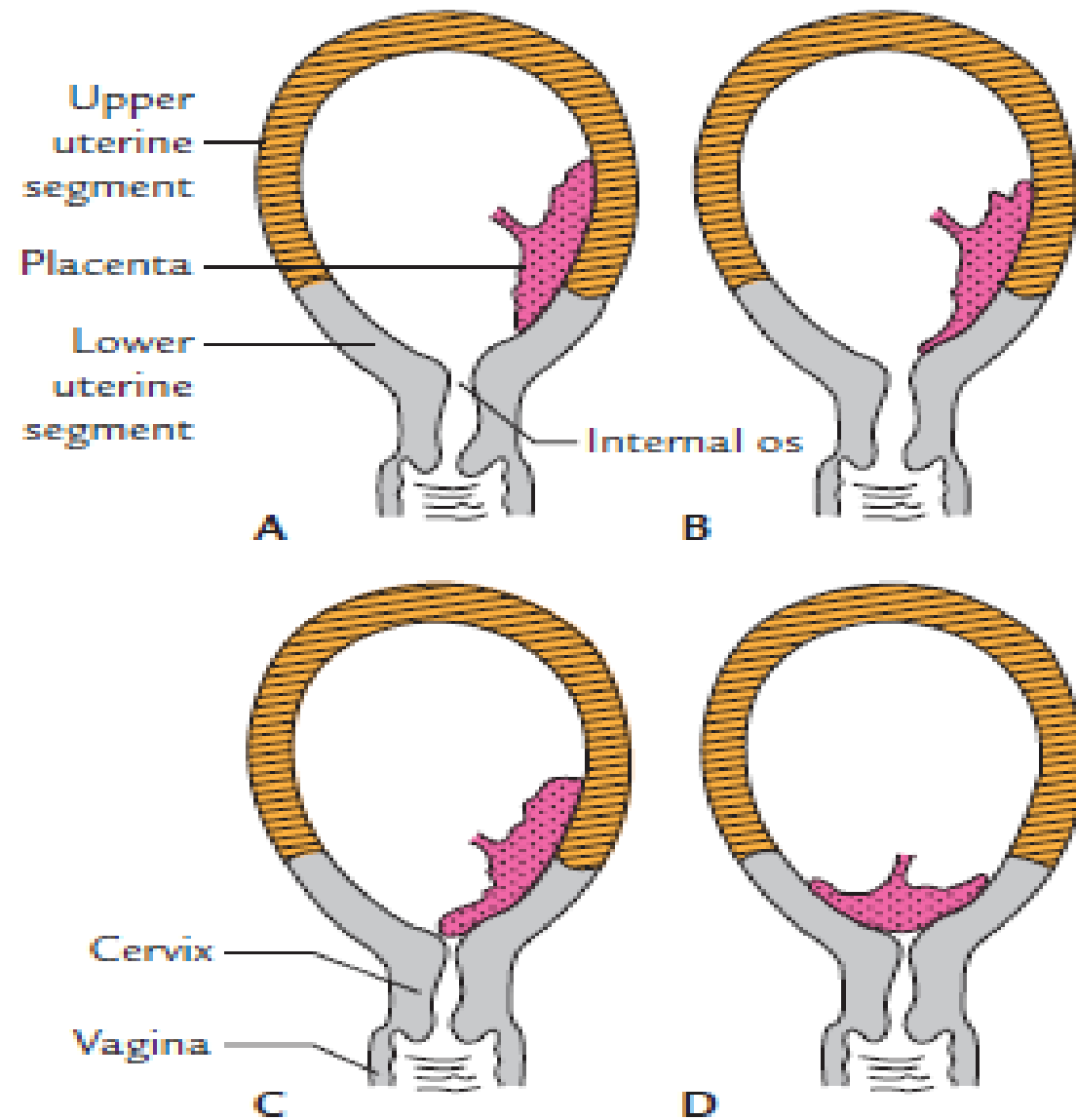


Fig. 6.7 Types of placenta previa. A. First degree. B. Second degree. C. Third degree. D. Fourth degree.

- ***Amniotic fluid***

- It is a clear, watery fluid (98%) and contains 2% solids (inorganic salts, urea, proteins, sugars)
- — Source: Fetal/maternal/both
- ◆ Amnion, fetal kidney, fetal lung, placenta
- ◆ Amnio-fetomaternal exchange
- — Amount
- ◆ 10th–20th week: 25–400 mL
- ◆ Increases up to 6th month, then decreases. At 28 weeks, it is 800 mL and at term it is 700 - 1000 mL

- ***Polyhydramnios***

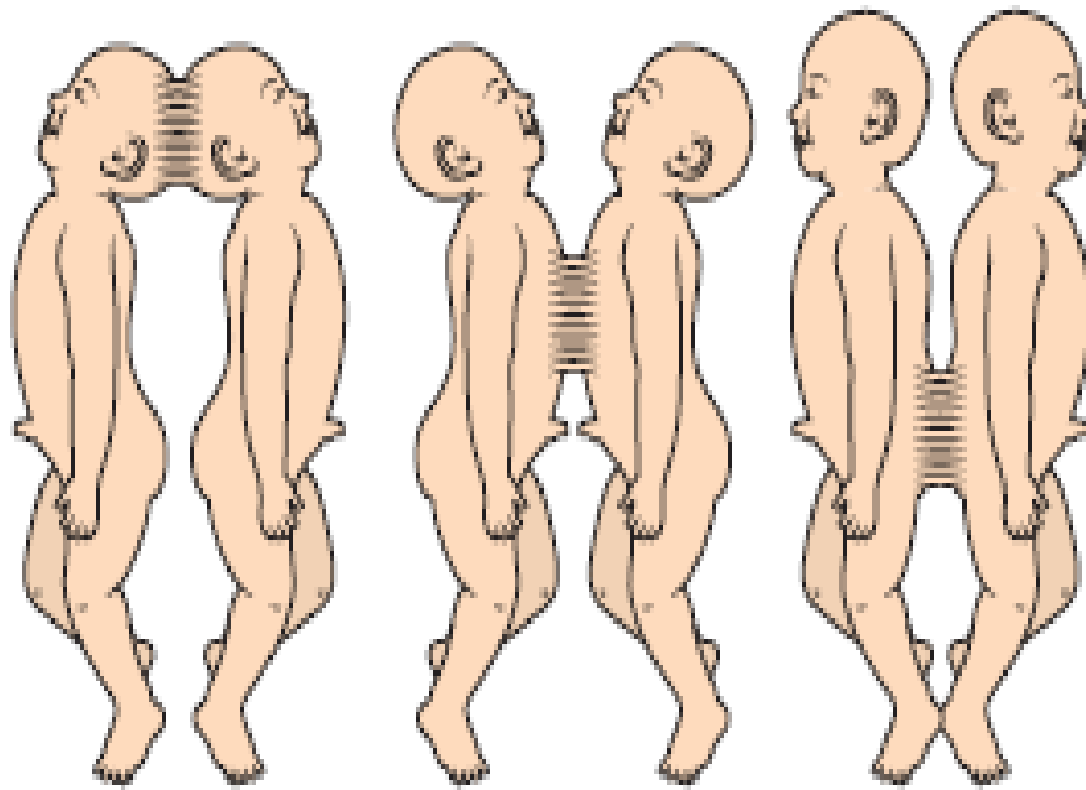
- excessive accumulation of amniotic fluid (e.g., 2000 ml or more) in the amniotic cavity leads to a clinical condition called *polyhydramnios*.
- — Abnormal production
- ◆ Hydramnios—more than 2 L of amniotic fluid will be present.
- In some cases, hydramnios is associated with atresia of the esophagus, or defects of central nervous system
- (CNS) which prevents swallowing of amniotic fluid by the fetus

- ◆ Oligoamnios/***Oligohydramnios***—scanty amniotic fluid.
- Normal amount of fluid at full-term is 700–1000 ml. If it becomes 400 ml or less, it is termed oligohydramnios
- causes placental insufficiency with reduced placental blood flow, and loss of amniotic fluid due to preterm rupture of amnion
- It is sometimes associated with renal agenesis, as no urine is added to the amniotic fluid
- Both conditions can cause difficulties during childbirth

- ***Clinical importance of amniotic fluid***
- ***Amniocentesis:***
 - It is a technique to collect amniotic fluid diagnostic purposes
 - fluid is collected either through cervix or anterior abdominal wall
 - procedure is usually done during 15–20 weeks of pregnancy
 - There is risk of fetal injury or preterm delivery in performing this procedure
 - **done:**(a) to examine the chromosomes in cells of amniotic fluid for detection of genetic diseases such as Down's syndrome
 - (b) to detect defective enzyme involved in the formation of myelin sheath in *Tay-Sachs* disease by biochemical techniques
 - (c) to detect neural tube defects.

- **sex of a fetus** can also be detected by chromosomal studies
- fetal urine is added to the amniotic fluid, it can be used to study fetal enzymes and fetal hormones.
- Presence of **high level of α -fetoprotein in amniotic fluid indicates neural tube defects**
- ***Chorionic villus biopsy:***
- *Small sample of chorionic tissue is removed under the guidance of ultrasound*
- done to detect the genetic disorders much earlier than the amniocentesis permits
- Carried out from weeks 10-11

- ***Incomplete duplication of disc***
- This results in the formation of conjoined twins or double monsters or Siamese twins
- Craniopagus—twins united at head
- Thoracopagus—twins showing fusion of thorax
- Pygopagus—fusion at sacral region
- Cephalothoracopagus—fusion of thorax and head



Craniophagus Thoracophagus Pygophagus

Fig. 6.24 Types of conjoined twins.

2. *Parasitic twins*: When one member of conjoined twins remains rudimentary due to diminished blood supply and grows like a parasite from the body of the well-developed co-twin, it is termed parasitic twin (Fig. 6.25).

Occasionally the parasitic twin may be completely enclosed within the body of the co-twin. In this condition, the fetus is termed *fetus in fetu*.

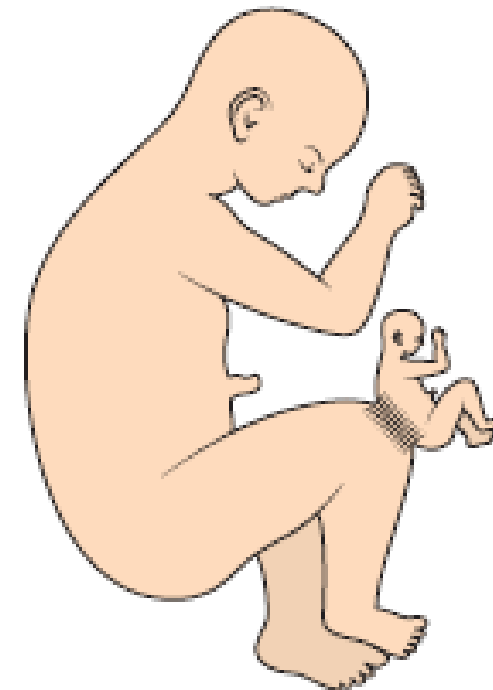


Fig. 6.25 Parasitic twins.